SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

X	Stopped engine
X	Shut off starting air supply – *At starting air receiver*
X	Block the main starting valve
X	Shut off starting air distributor/distributing system supply
X	Shut off safety air supply – *Not ME engines*
X	Shut off control air supply
X	Shut off air supply to exhaust valve – *Only with stopped lubricating oil pumps*
X	Engage turning gear
X	Shut off cooling water
X	Shut off fuel oil
X	Stop lubricating oil supply
X	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D01-05	Cylinder cover stud, check distance	128-129	mm
D02-01	Test pressure	7	bar
D02-02	Piston rod/crosshead, tightening torque	430	Nm
D02-03	Piston rod/crown, tightening torque	300	Nm
D02-05	Piston skirt, tightening torque	80	Nm
D02-06	Cooling oil pipe, tightening torque	80	Nm
D02-07	Piston rod/crosshead, tightening torque+angle	100/30	Nm/°
D02-08	Piston ring new, radial width	17.2	mm
D02-09	Piston ring worn, min. radial width	14.2	mm
D02-10	Groove No.1, max. vertical height	13.2	mm
D02-11	Groove Nos. 2, 3 and 4, max. vertical height	10.2	mm
D02-12	Piston top, max. permissible burn-away	15	mm
D02-13	Piston ring new, height of ring No. 1	12.4	mm
D02-14	Piston rings new, height of ring Nos. 2, 3 and 4	9.4	mm
D02-15	Minimum free ring gap (before dismantling)	28	mm
D02-16	Minimum ring gap, ring No. 1 (new ring in new liner)	3.4	mm
D02-17	Minimum ring gap, ring Nos. 2, 3 and 4 (new ring in new liner)	2.4	mm
D02-18	Vertical clearance, new parts	0.38	mm
D02-19	Vertical clearance, worn parts, max.	0.87	mm
D02-20	Piston complete	1000	kg
D02-21	Piston crown	220	kg
D02-22	Piston rod	580	kg
D02-23	Piston skirt	60	kg
D02-24	Piston cooling pipe	22	kg
D02-25	Lifting tool, tightening torque	100	Nm
D02-46	CPR ring CL groove, min. depth	1.1	mm

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90251	040	Lifting tool for piston rod foot
P90251	075	Template for piston top
P90251	087	Distance piece for stuffing box
P90251	099	Cover for stuffing box hole
P90251	110	Pressure test tool for piston
P90251	134	Piston ring expander
P90251	158	Guide screw for piston crown
P90261		Piston and Piston Rod - Tools
P90264		Tools for tilted lift (optional tool/If applied)
P90265		Piston and Piston Rod - Support Tools
P90266		Piston - Lifting Tools
P90451	118	Rubber cover for crosshead

Scavenge Port Inspection

To detect possible leakages from the piston or cylinder cover, keep the cooling water and cooling oil circulating during the scavenge port inspection.

1. The scavenge port inspection is carried out from the scavenge air receiver. An additional view of the rings is possible through the cleaning cover on the manoeuvring side.

   Turn the engine at least ½ a revolution, and begin with a unit arriving downwards, just above the scavenge air ports. Inspect the piston rod and the lower part of the cylinder wall.

   While the piston is passing downwards, inspect the piston skirt, all the piston rings, the ring lands and the piston top.

2. Ring inspection:
   Inspect the rings, one at a time, and note down the results. *See Volume I, Operation, Chapter 707.*

3. Ring tension:
   Check the tension of the piston rings, by pressing against them with a wooden stick.

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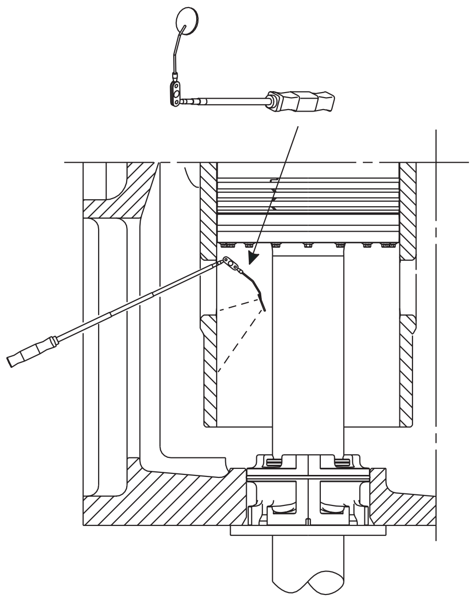

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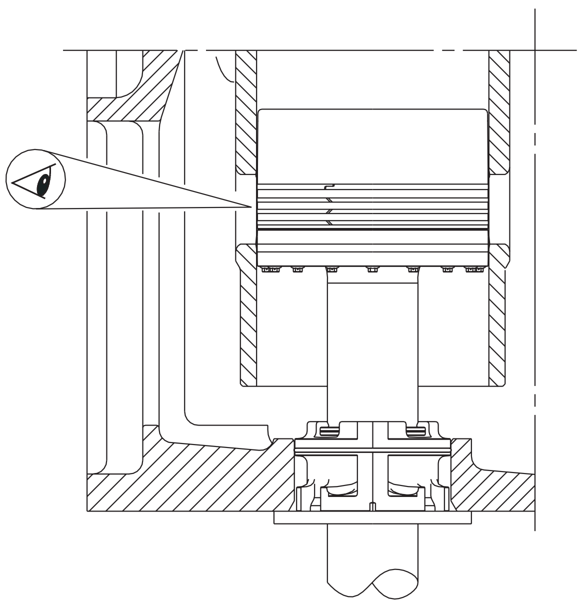

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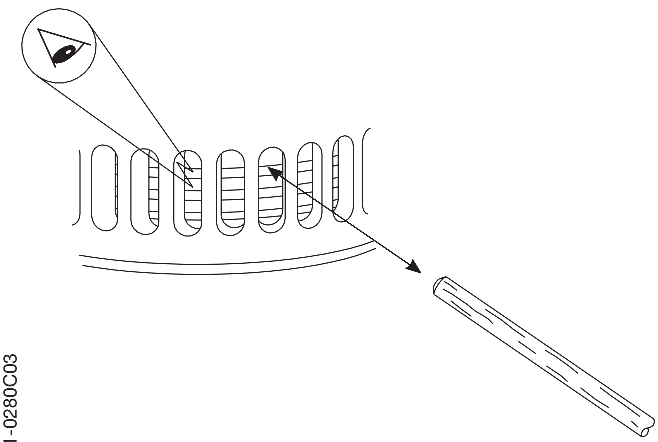

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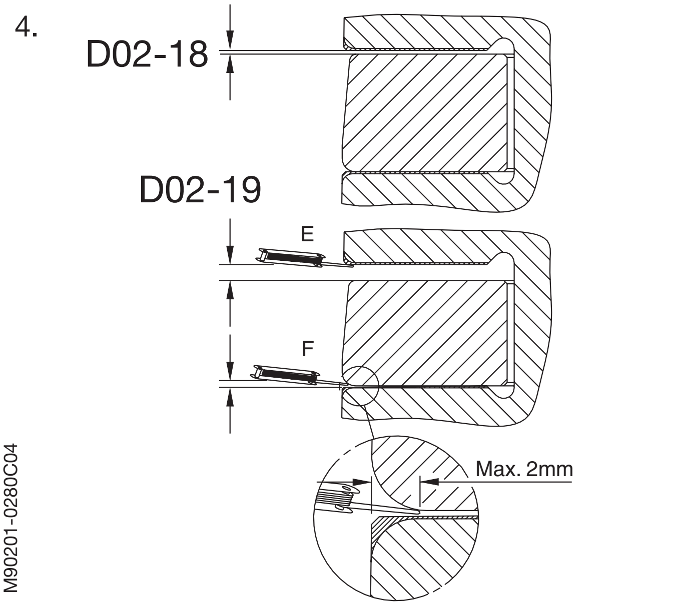

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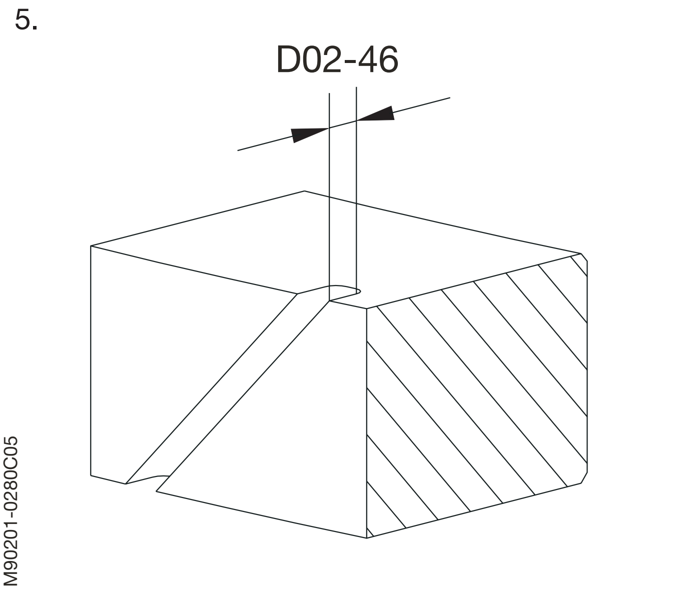

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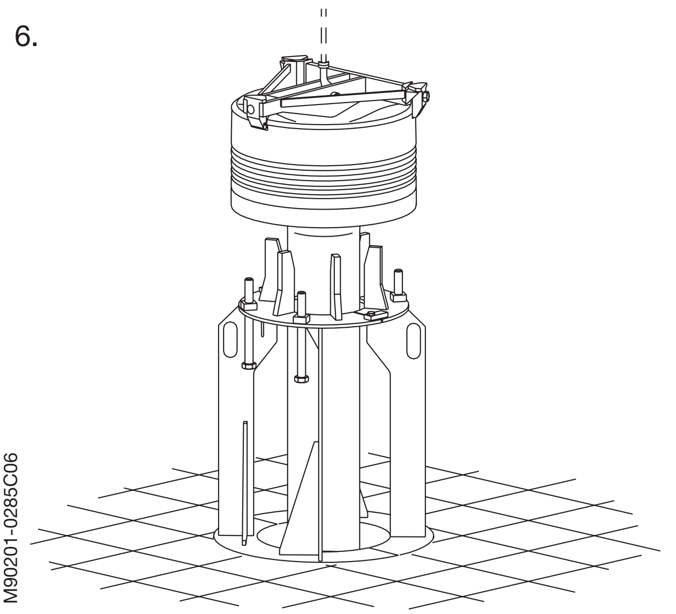

4. Ring grooves:
   Measure the total clearance between the piston ring and the ring groove. The total clearance *must not* exceed the value stated in Data.

   Measure the clearance at the top (E) and bottom (F) of the piston ring groove.
   Total clearance = E + F.

5. Uppermost piston ring:
   If possible, measure the depth of the pressure relief grooves with a calliper. The piston rings must be replaced if the radial depth of the grooves has worn down to less than stated in Data D02-46.

Checking, in connection with piston overhaul

6. Piston support:

   Remove the piston from the cylinder and place it on the piston support.
   See Procedure 902-1.2. For evaluation of the rings, see Volume I, Chapter 707.

Note!
It is recommended to replace all the piston rings whenever a piston is removed from the engine.

7. Free ring gap:
   Before dismantling the piston rings, measure the free ring gap of all the piston rings.

   For dismantling the piston rings, see Procedure 902-1.3.

8. Radial ring width:
   Measure the radial width of the rings. Note down the results *See Volume I, Operation, Chapter 707.*

9. Ring grooves:
   Clean the ring grooves and check them for burn marks or other deformation.

   Measure the ring grooves with a calliper gauge, *see Data D02-10 and D02-11*.

   Clearance in piston ring grooves:
   The maximum vertical height in a worn ring groove **must not** exceed the value stated in Data. The groove is also worn out if there is no chromium layer.

   If the ring grooves are worn out, the piston crown must be reconditioned, contact MAN B&W Diesel A/S for advice.

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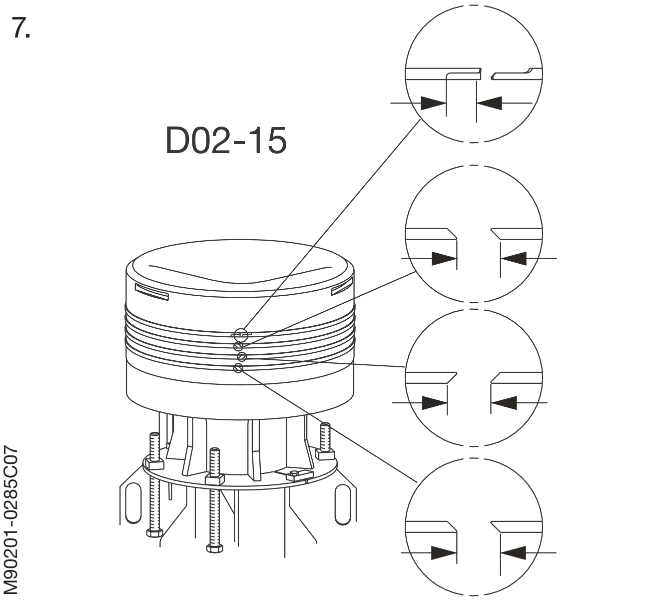

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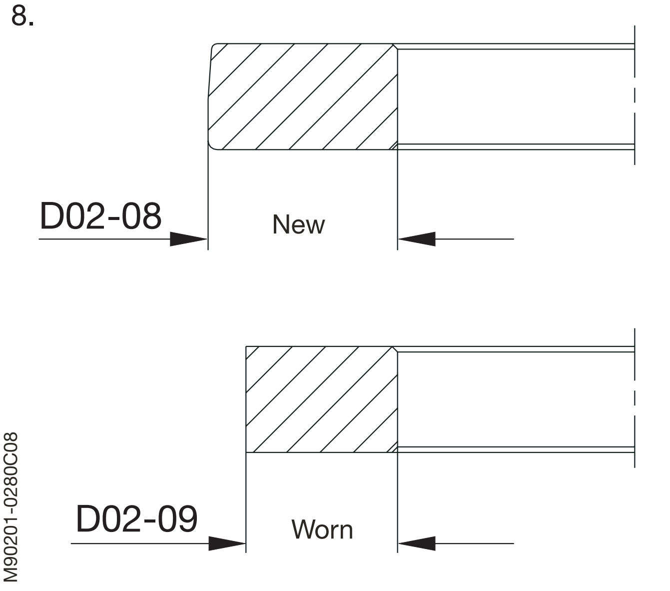

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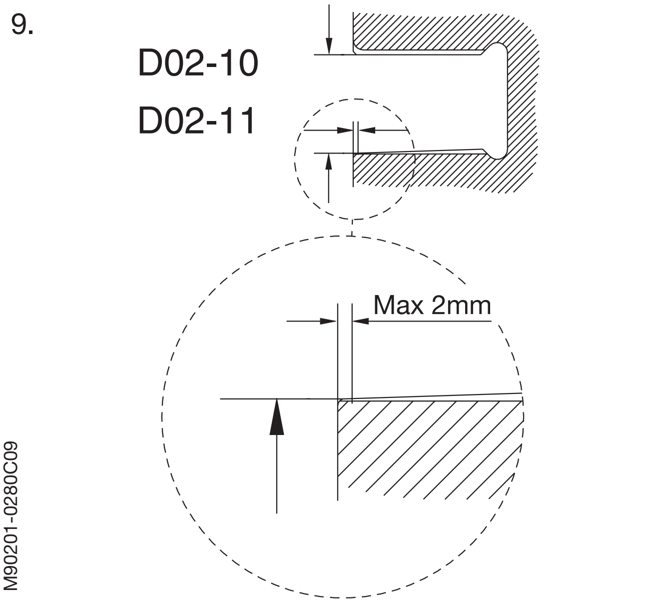

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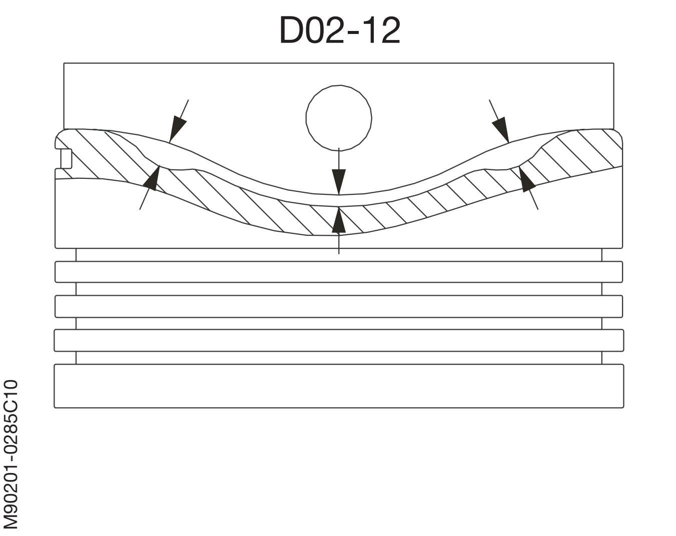

10. Piston crown top:
    Clean the piston crown and check the burn-away by means of the template.

    For maximum permissible burn away value, see Data.

    Check the burn-away on the whole circumference of the piston crown top.

    If the burn-away exceeds the values given in Data, contact MAN B&W for advice.

    Note down the results for later reference.

Preparations in the crankcase

1. Access:
   Open the crankcase door to the cylinder concerned.

   Turn the crosshead down far enough to give access to the piston rod stuffing box and the piston rod/crosshead connection.

2. Stuffing box:
   Release the stuffing box by removing the innermost screws from the stuffing box flange.

   On engine models where the drain pipe is connected directly to the stuffing box:

   Disconnect the stuffing box drain pipe.

Note!
Do **NOT** remove the outermost screws from the flange.

3. Loosen the piston rod-crosshead connection:
   Remove the screws from the piston rod.

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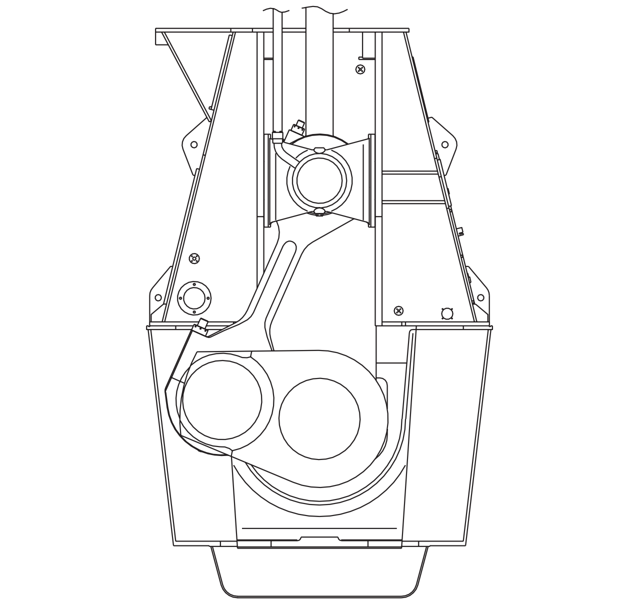
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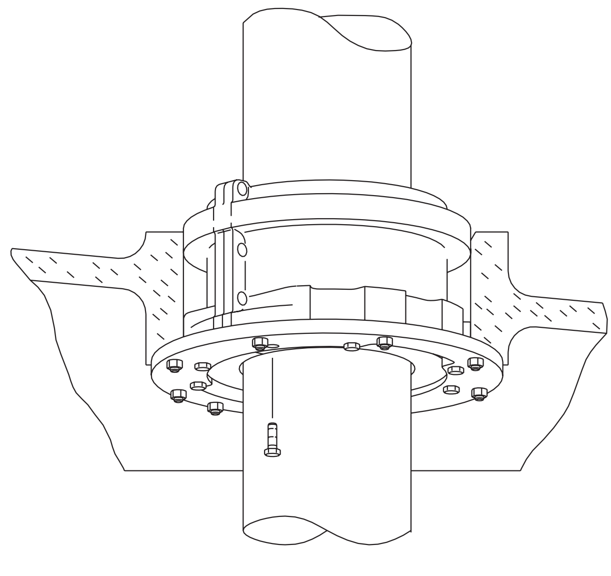
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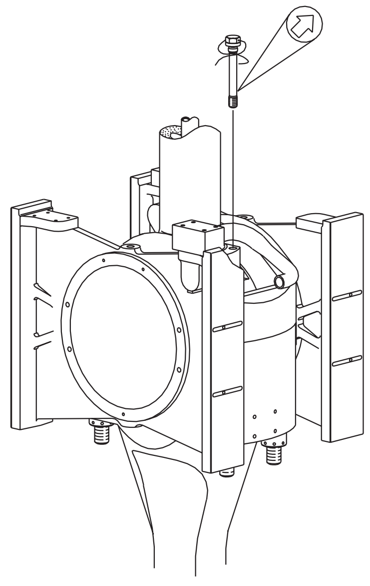
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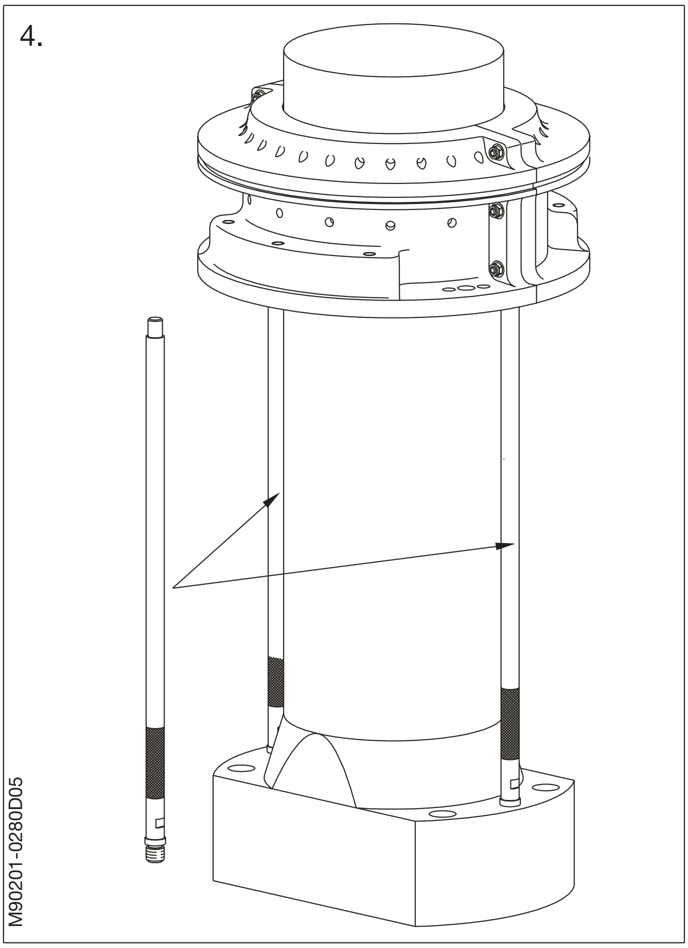
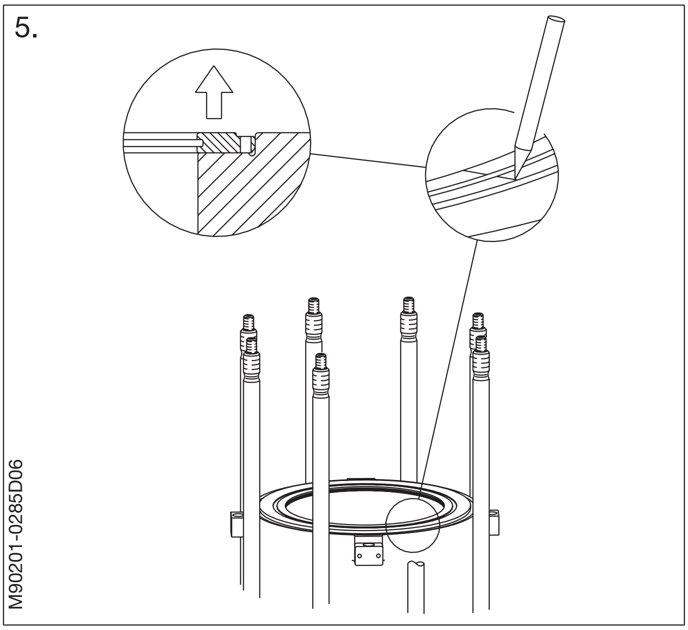




4. Stuffing box distance pieces:
   Mount the two distance pieces on the piston rod foot to protect the lower scraper ring and to guide the stuffing box.

Preparations on the cylinder top

5. Cylinder cover:
   Remove the cylinder cover. *See Procedure 901-1.2.*

   Make a scratch mark in liner and piston cleaning ring to ensure correct re-mounting. Remove the piston cleaning ring.

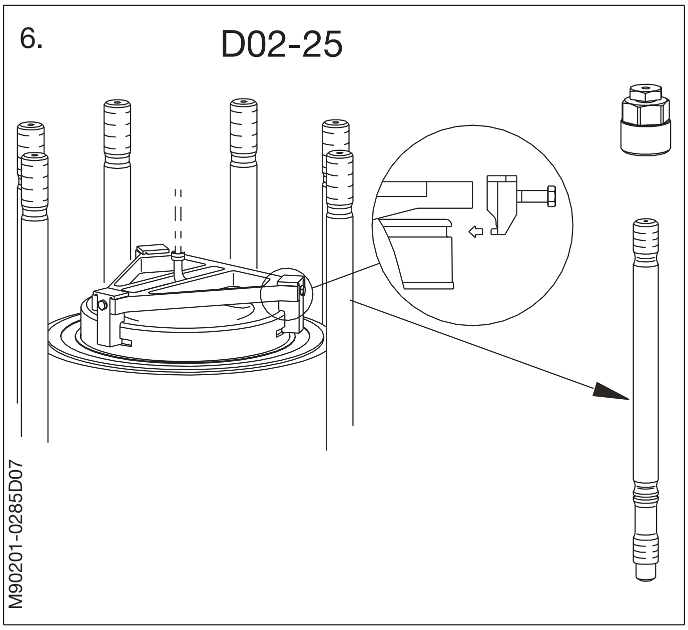
   Carefully remove any wear ridges at the top of the cylinder liner. *See Procedure 903-1.3.*

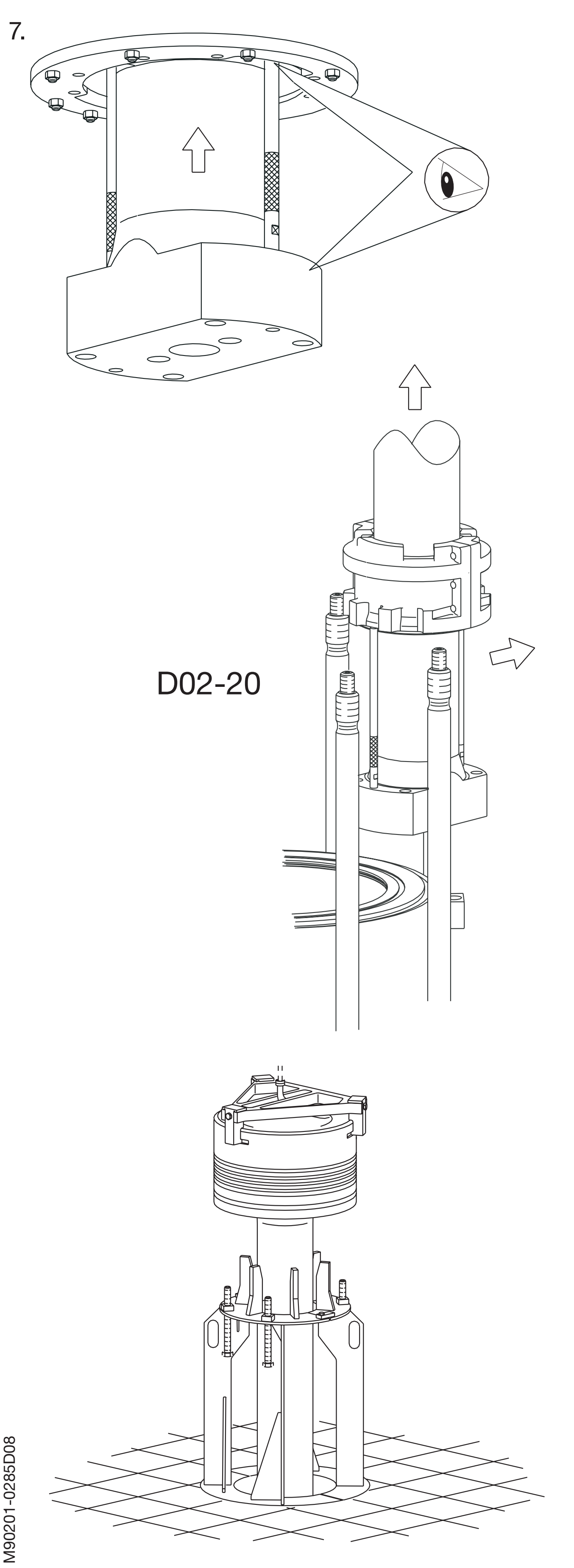
6. Piston lifting tool:
   Turn the piston to TDC. The top of the piston is now free of the cylinder liner.

   Clean the lifting groove of the piston crown and mount the lifting tool.

Note!
Make sure to mount the lifting tool correctly, so that the claws of the lifting tool enter the lifting grooves of the piston crown.

   If the engine not is equipped with long distance pieces, remove one or two cylinder cover studs, using a stud setter.





Piston lift

7. Lift the piston out of the cylinder liner and guide the piston rod foot through the stuffing box flange.

   If the engine is equipped with long distance pieces for the stuffing box, the piston rod foot can pass between two cylinder cover studs.

   Place the two halves of the support around one of the openings in the platform. Lower the piston rod foot and stuffing box through the opening in the platform. Secure the two support halves with screws and lower the piston on the support. Check the piston is resting on the piston rod flange.

   Limited Lifting height:

   If the piston rod foot can not be lifted clear of the liner top with the standard lifting tool, contact MAN Diesel for instructions for tilted lift.

8. Protect the crosshead bearing:

   Place a cover over the opening for the piston rod stuffing box in the bottom of the cylinder unit.

   Turn the crosshead down far enough to permit mounting of the protective rubber cover on the crosshead bearing cap. The protective rubber cover is found on tool panel 904.

   The covers must remain in place to protect the crosshead bearing journal from impurities until the piston is remounted.

   Clean, measure and recondition the cylinder liner. *See Procedure 903-1.*

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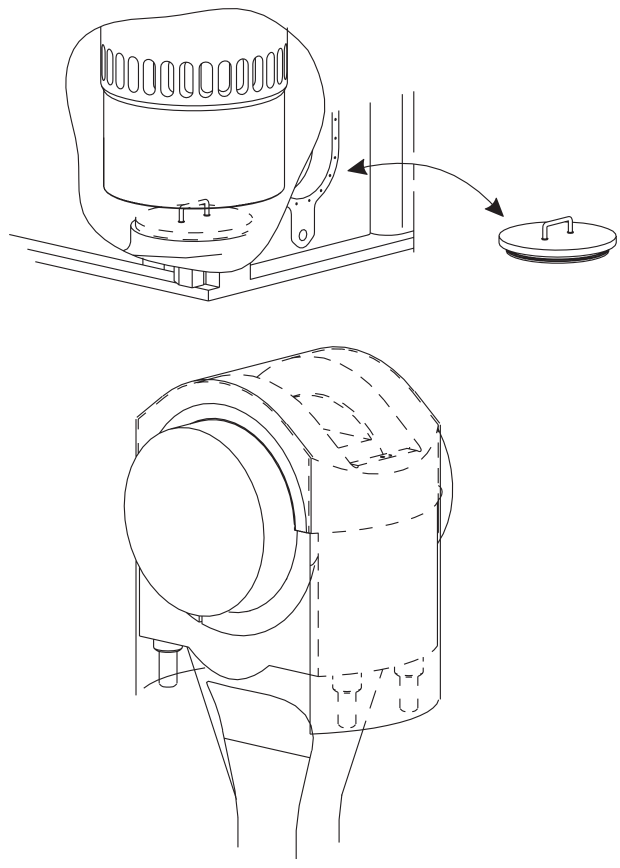

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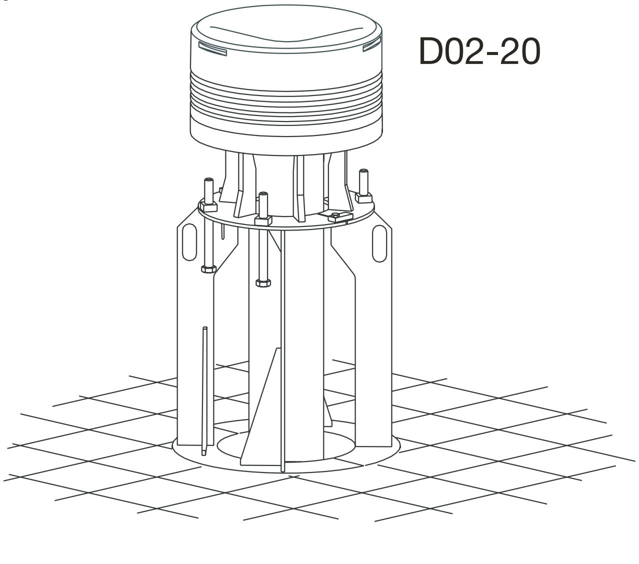

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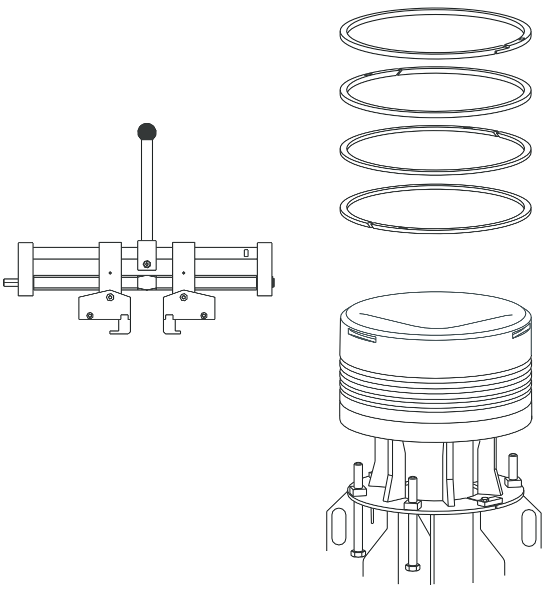

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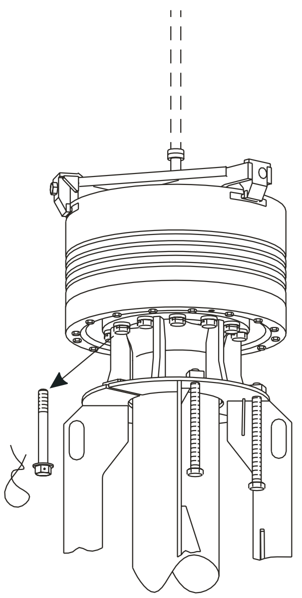


1. Piston support:
   Place the piston in the support and remove the piston lifting tool.
   See Procedure 902-1.2.

   Clean the piston top and the piston rings.

   Check the free ring gap and the burn-off on the piston top. *See Procedure 902-1.1.*

   Remove the stuffing box. *See Procedure 902-2.3.*

2. Piston ring dismantling:
   Take off the piston rings by means of the ring expanders. If the engine is equipped with two ring expanders, the short ring expander is for the uppermost ring.

   First remove the uppermost ring, then ring No. two, three and four.

   Clean and inspect the rings and the ring grooves. *See Procedure 902-1.1.*

3. Piston crown dismantling:
   Remove the locking wire and the innermost screws between the rod and the piston crown.

   Lift the piston crown and skirt clear of the piston rod.

4. Piston rod and cooling pipe:
   Remove the screws from the cooling oil pipe flange.

   Mount the eye bolts and lift out the cooling oil pipe.

   Clean and inspect the cooling oil pipe and the piston rod, then remount the cooling oil pipe, *see Data.*

   Check that the surfaces of the O-ring groove is clean and smooth.

   Mount a new O-ring on the piston rod flange.

5. Unhook the crane from the piston lifting tool, and connect the lifting tool and the crane hook by means of a wire rope. Lift up the piston crown.

   Fit two eyebolts in the piston skirt and a wire rope between the bolts as shown in the sketch.

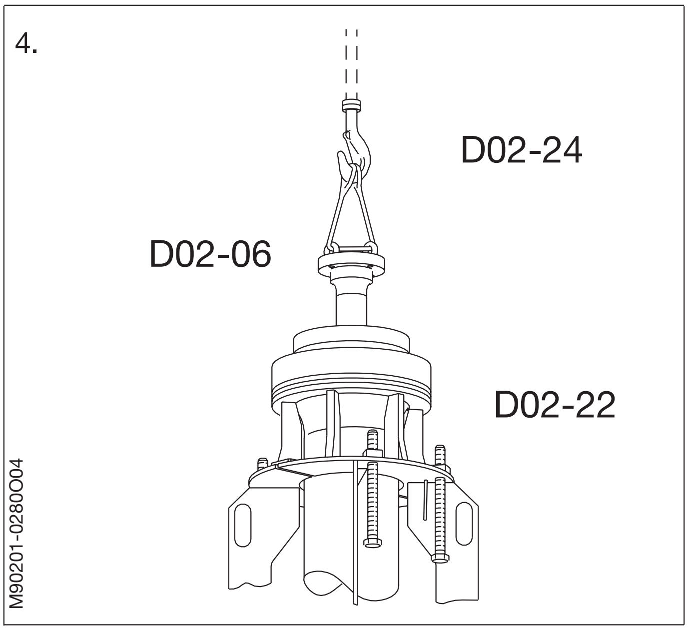
   Install a tackle in a suitable place with sufficient space below for the piston crown. Attach the tackle to the wire rope on the piston crown and carefully turn the piston crown upside down. Use both cranes if the engine room is equipped with two cranes.

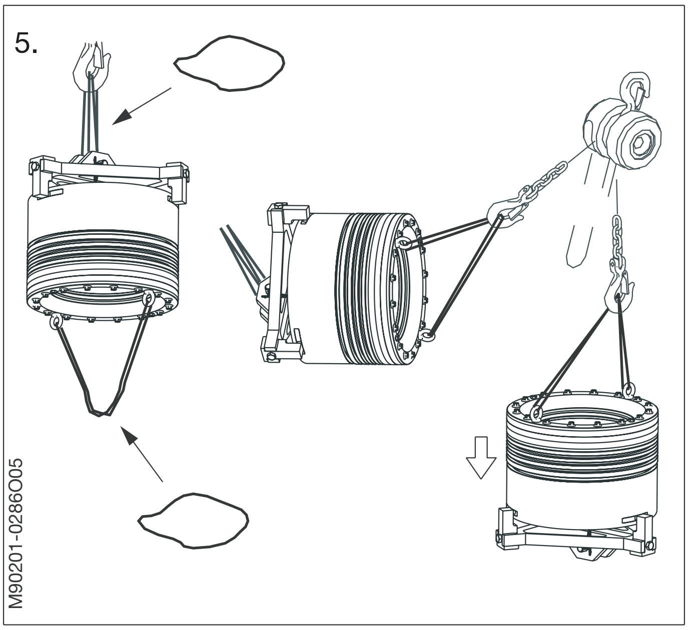
   Land the piston lifting tool and the piston crown on a sufficient support of wood pieces. Loosen the piston lifting tool and lift the piston crown clear of the tool.

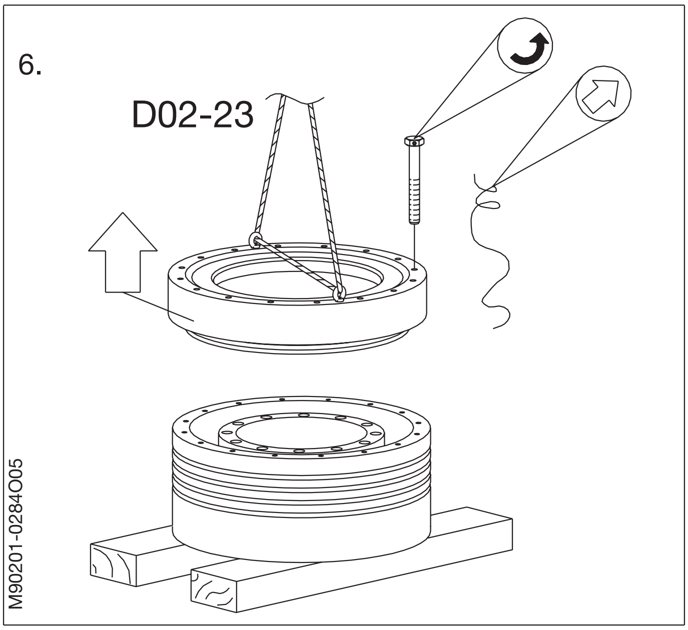
6. Piston crown dismantling and cleaning:
   Place the piston crown with skirt on a wooden support as shown.

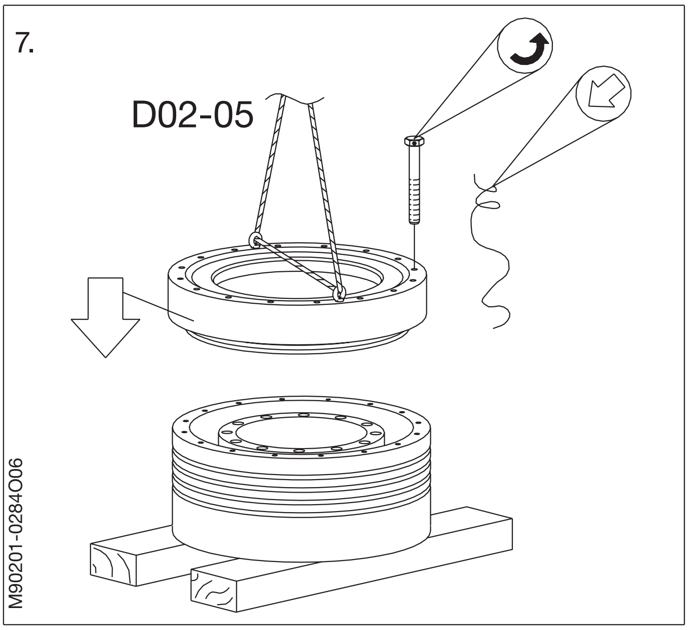
   Remove the locking wire and the screws in the skirt. If necessary use two dismantling screws to pull the skirt out of the piston crown. Mount two eye bolts in the skirt. Lift the skirt and land it on a couple of planks.

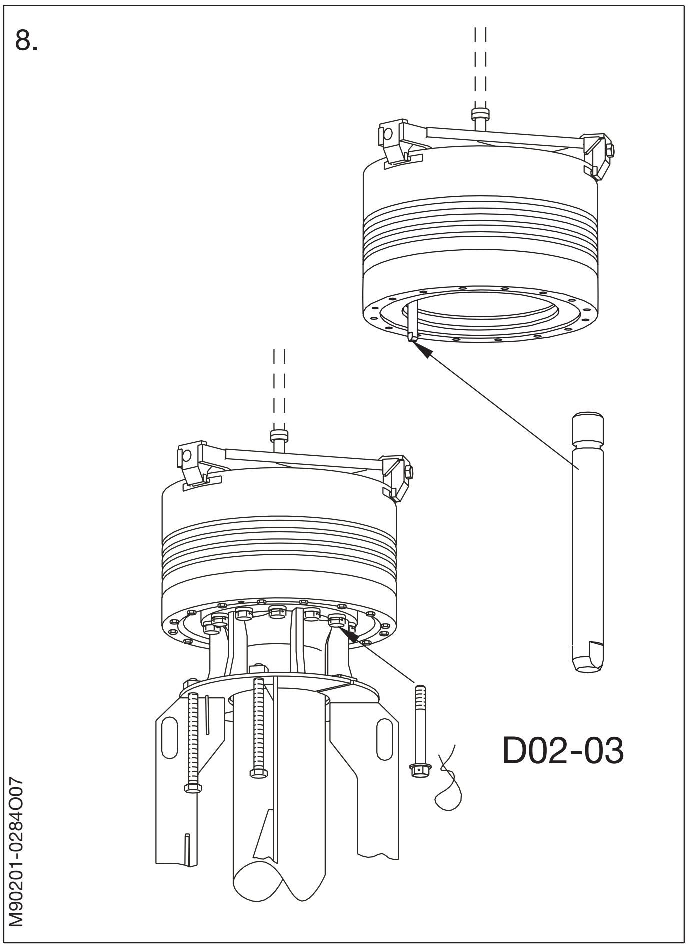
   Discard the sealing ring on the piston skirt.











Thoroughly clean and inspect all parts of the crown and skirt. If coke deposits are found in the cooling spaces of the piston crown, they should be washed clean with Carbon Remover or a similar cleaning fluid. When all coke deposits have been dissolved, clean and inspect the piston crown again.

Note!
Coke deposits reduce heat transfer from the piston crown to the cooling oil. The deposits must be removed as a routine procedure when a piston is overhauled.

7. Piston crown assembly:
   Mount a new O-ring on the piston skirt. Check that the surfaces of the O-ring groove are clean and smooth. Coat the ring with lubricating oil before mounting.

   Mount the piston skirt on the piston crown. Tighten the screws to specified torque, *see data D02-05.* Lock the screws with locking wire, *see Procedure 913-7.*

8. Piston crown and skirt mounting:
   Mount the guide screw to ensure the correct positioning of the piston rod in relation to the piston crown.

   Lubricate the O-ring on the piston rod flange with lubricating oil.

   Land the overhauled piston crown and skirt on the piston rod.

   Remove the guide screw.

   Mount and tighten the screws between the piston rod and the piston crown, *see Data.*

Sealing ring and pressure test

According to current class rules, the piston must be pressure tested hydraulically. It is possible to carry out a test of the sealing rings with compressed air before filling the piston with oil. The sealing ring test can also be carried out when the piston is resting in the support tool.

9. Sealing ring test:
   Mount the pressure-testing tool on the piston rod foot. Connect compressed air to the testing tool and fill the piston to 4-5 bar. Close the valve on the testing tool and remove the air connection. The piston must now hold the pressure for minimum 30 minutes.

   Spray a little soap water on to the surface joints between piston rod/crown/skirt and around the bolt heads to detect leaks.

   Dry off all soap water.

10. Pressure test:
    For this test, the piston must be turned upside down *(see next step to turn a complete piston).*

    Fill the piston and piston rod with lubricating oil. Mount the pressure-testing tool on the piston rod foot. Pressure-test the piston at the pressure stated on the Data Sheet. Check the contact surfaces of the piston and the sealing rings for tightness. Check that there are no cracks in the piston crown.

    Turn the piston upside up and drain out the piston oil.

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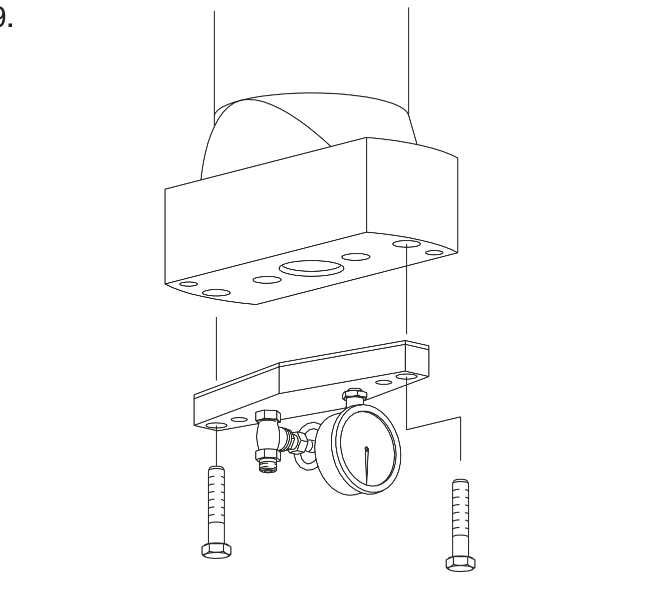

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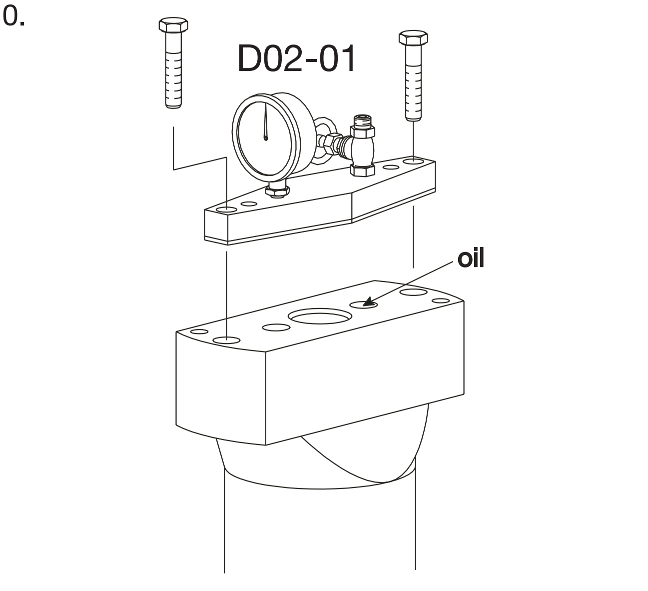

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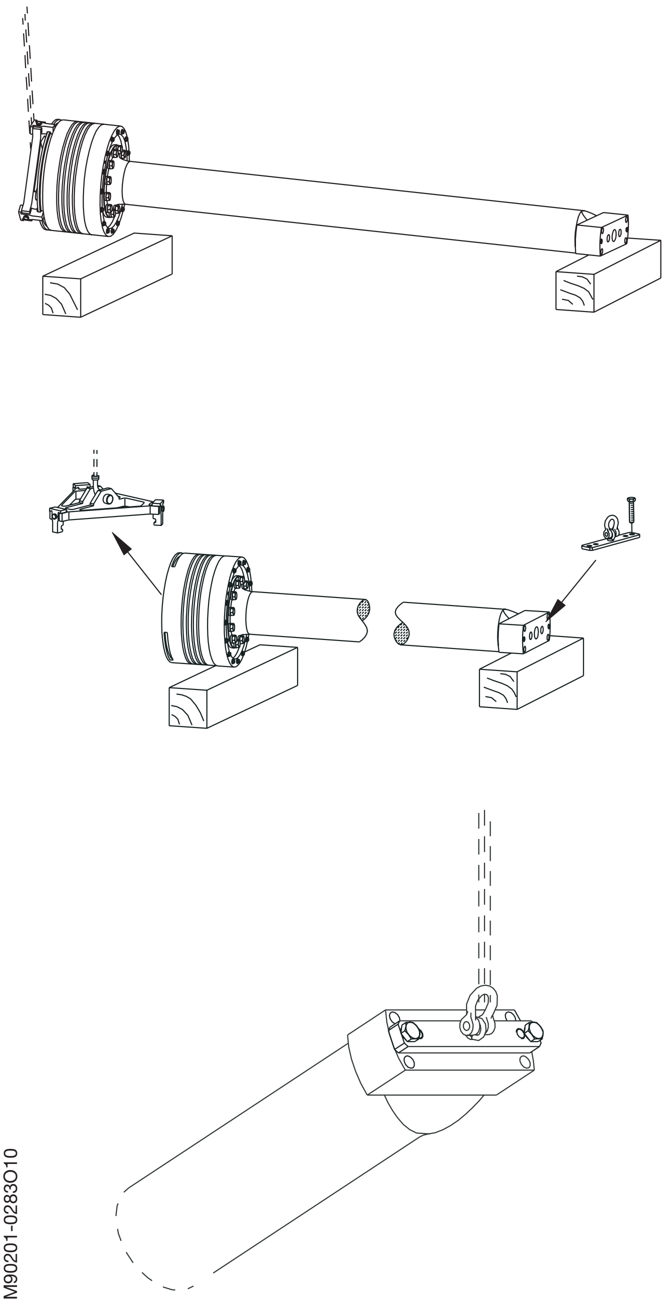

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11. Piston turning:
    Lift the piston with the normal lifting tool. Lower the piston rod foot until it is close to the platform. Land the foot on a wooden block.

    Lower the piston crown to the platform and land it on a wooden block in such a way that it is possible to remove the lifting tool.

    Attach the lifting bracket to the bottom of the piston rod foot. Hook the crane on to the lifting bracket. Lift the piston rod foot clear of the wooden block. Keep lifting until the piston rod is in a vertical position.

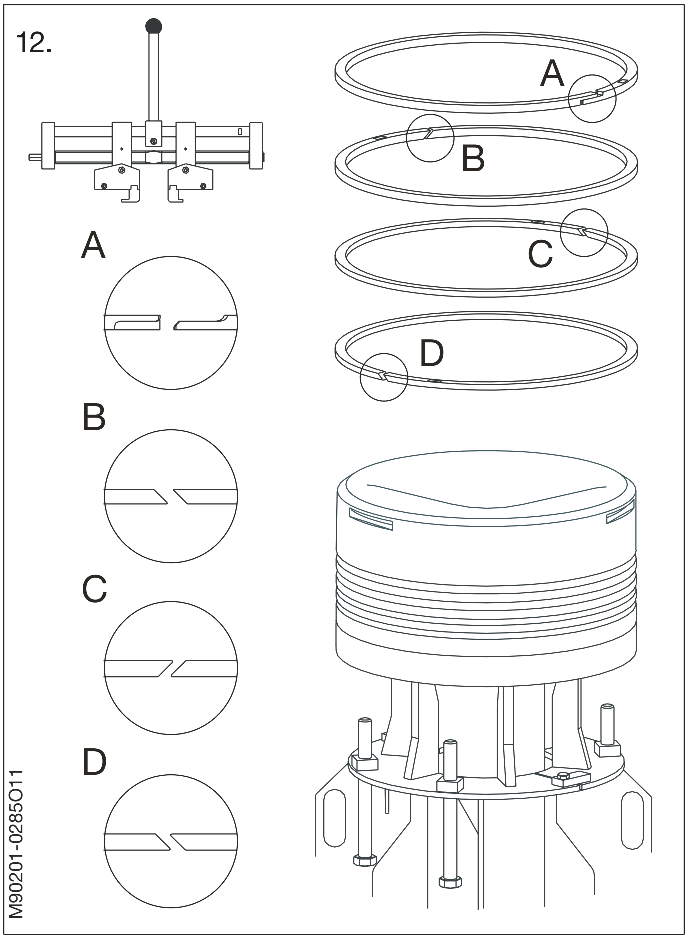
Note!
During the lift, follow with the crane to keep the crane positioned vertically above the lifting point. The stuffing box must be removed.

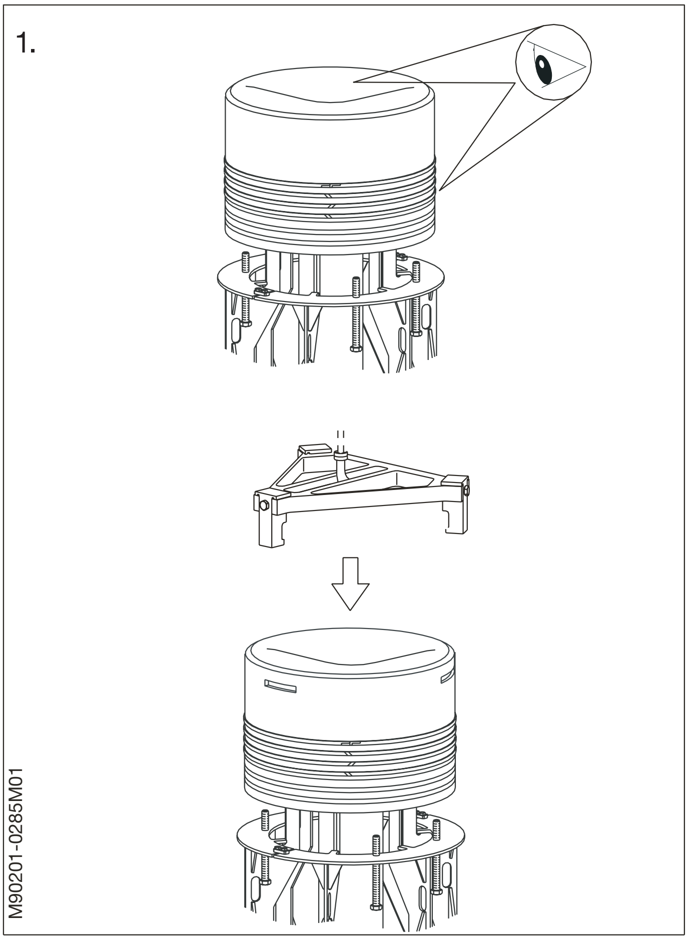
12. Piston completion:
    Fit the new piston rings (alternately right-hand and left-hand cuts, with the ring gaps staggered 180° and with the TOP mark upwards), using the ring expander.

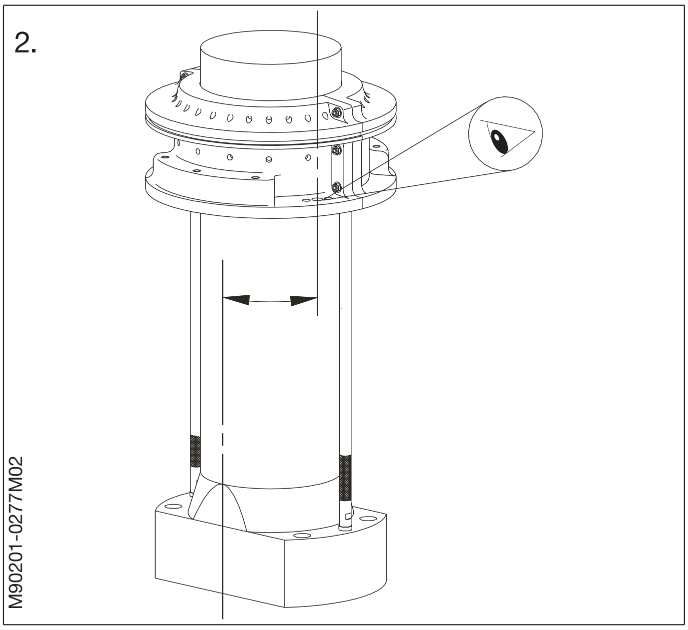
    When mounting the piston rings, use the ring expanders to prevent unintended deformation of the rings.

    Do not expand the rings more than necessary. The uppermost ring (CPR-ring) must be mounted with the short ring expander, if the engine is equipped with two ring expanders

    Mount the piston rod stuffing box.
    See Procedure 902-2.3.







Preparation of piston

1. Check the piston rings and piston crown in accordance with Procedure 902-1.1, if not already done.

   Mount the lifting tool on the piston crown. *See Data D02-25.*

2. Stuffing box position:
   Ensure that the stuffing box is correctly positioned over the distance pieces mounted on the piston rod foot. Both the holes for the flange and the drain hole for the drain pipe must be positioned correctly.

Preparation of cylinder liner

3. Mount the guide ring:
   Mount the guide ring in the top of the cylinder liner. The cut outs for the lifting tool must be turned to fit the piston lifting tool.

4. Stuffing box cover:
   Remove the cover from the piston rod stuffing box opening in the bottom of the cylinder unit.

   Clean the stuffing box flange.

5. Crosshead position:
   Turn the crosshead to a position 45° from TDC (crank web pointing towards exhaust side).

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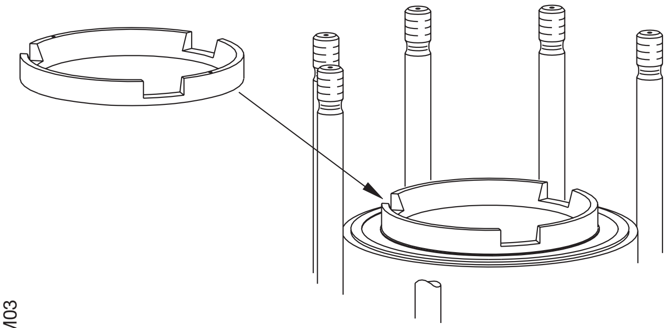

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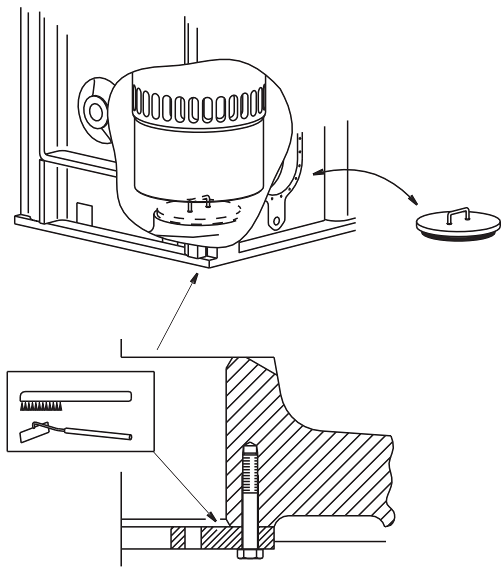

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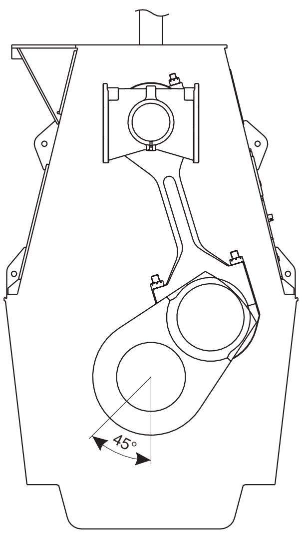

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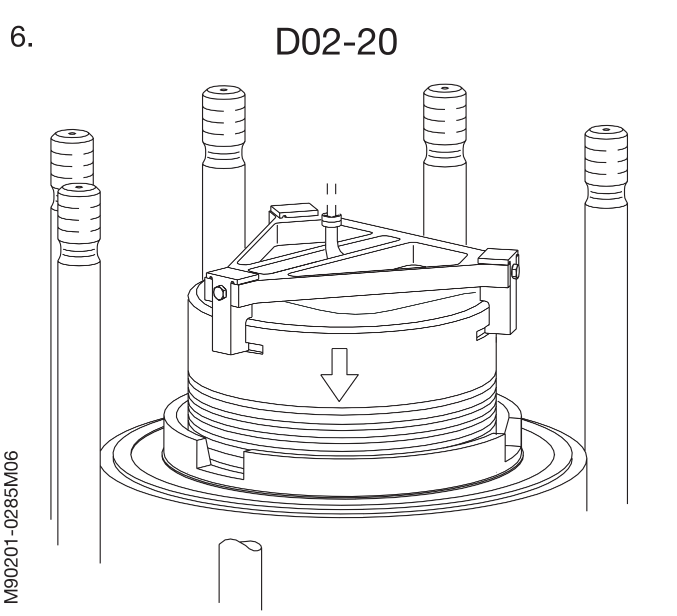

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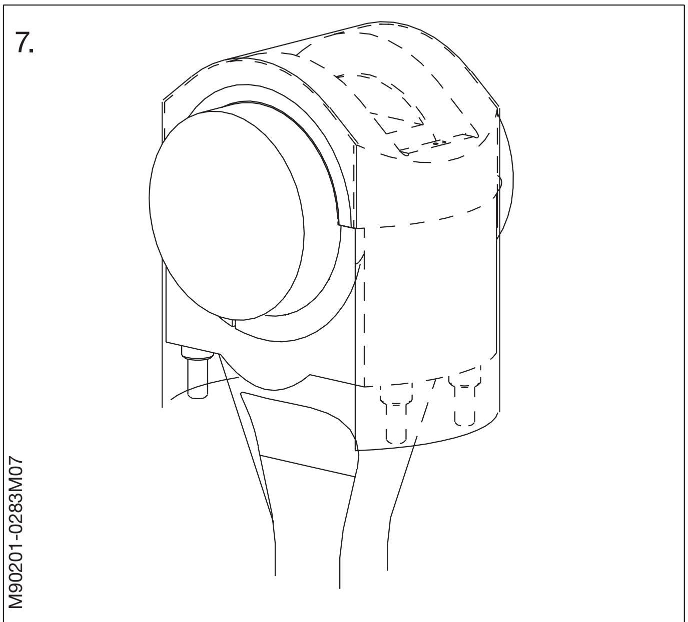

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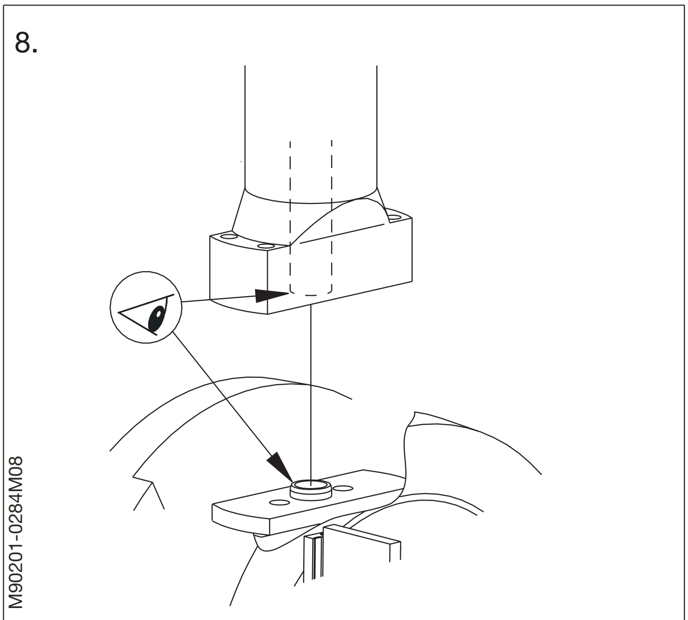


Mounting of piston

6. Coat the O-rings of the stuffing box and the piston rod with oil. Coat the piston rings and cylinder liner with cylinder lubricating oil.

   Lower the piston into the cylinder liner – while guiding the piston rod foot through the cut-out in the stuffing box flange – until the piston rings are inside the liner.

7. Protective cover:
   Remove the protective rubber cover from the crosshead.

8. Crosshead alignment:
   Turn the crosshead almost to TDC, while checking that the guide ring of the crosshead enters the centre hole in the piston rod.

   After turning the crosshead fully to TDC, and ensuring that the piston rod has full contact with the crosshead, unscrew the lifting tool and remove the lifting tool and the guide ring for piston rings.

9. Stuffing box:
   Turn down and land the stuffing box on the stuffing box flange. Check that the holes in the stuffing box and stuffing box flange are correctly centered.

   Tighten the piston rod stuffing box by means of the screws through the inner holes in the stuffing box flange. *For Data, see Procedure 902-2.*

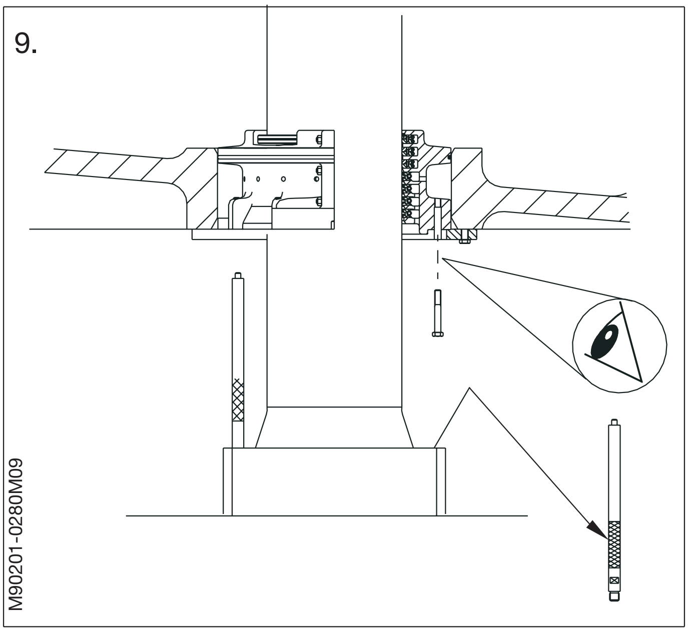
   On engine models where the drain pipe is connected directly to the stuffing box:

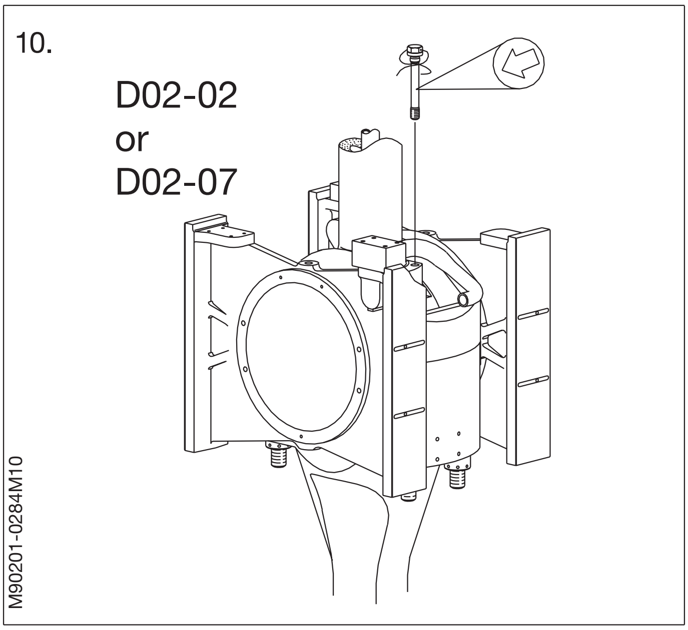
   - Mount the stuffing box drain pipe.

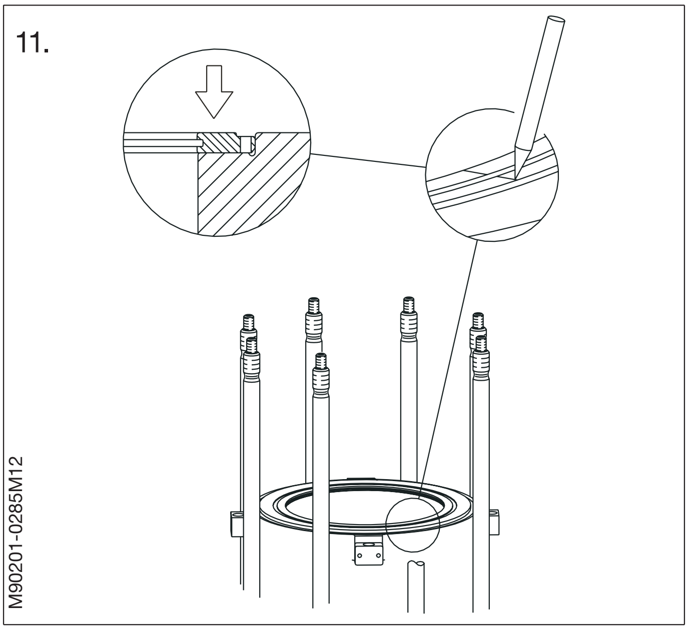
   Remove the distance pieces from the piston rod foot.

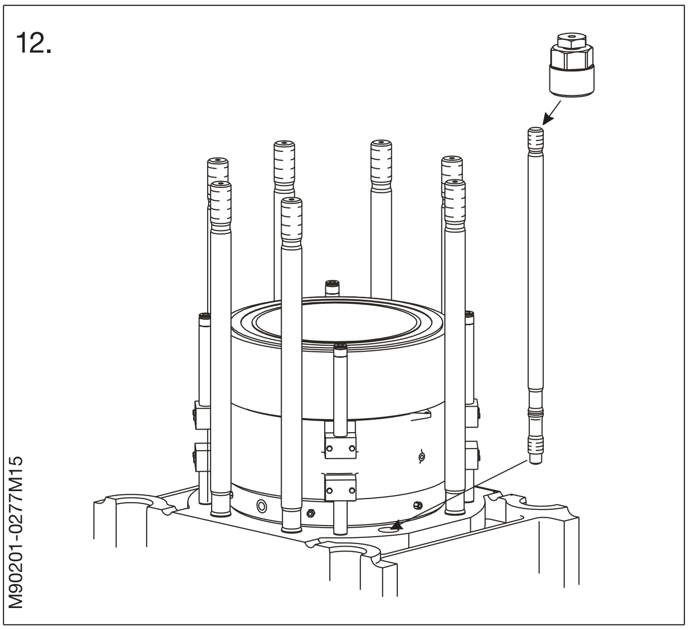
10. Tightening of the piston rod-crosshead connection:
    Mount and tighten the piston rod screws. Tighten the screws to the specified torque and lock with locking wire.
    See Data. Use either Data D02-02 or Data D02-07.

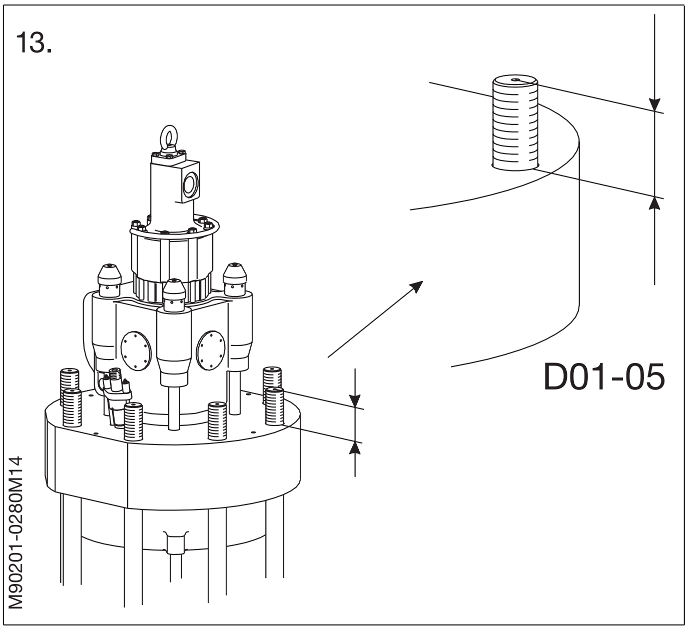
11. Piston cleaning ring:
    Mount the piston cleaning ring in accordance with the scratch mark. If the PC-ring is damaged (broken or cracked), it must be replaced by another ring.
    See Procedure 903-1.











12. Cylinder cover studs:
    If the cylinder cover studs have been removed, remount them. Carefully clean the surfaces around the base of the studs and check the O-rings on the studs.

    Mount the cylinder cover studs with the stud setter. Screw the stud down to contact and half a revolution back.

13. Cylinder cover:
    Land the cylinder cover on the liner and check the distance that the stud is protruding from the cylinder cover. If necessary, adjust the distance D01-05 by turning the stud.

    Tighten the cylinder cover nuts and mount the necessary pipes.
    See Procedure 901-1.4.

14. Running-in:
    Smear the piston rod with molybdenum disulphide, and turn the crankshaft a couple of revolutions.

    At the first opportunity, start the engine and keep it running for about 15 minutes at a speed corresponding to “Dead Slow” Ahead.

    Then stop the engine and inspect the piston rod and stuffing box.

SAFETY PRECAUTIONS

X	Stopped engine
X	Block the starting mechanism
X	Shut off starting air supply
X	Engage turning gear
X	Shut off cooling water
X	Shut off fuel oil
X	Shut off lubricating oil
	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D02-26	Stuffing box flange, outer screws tightening torque	80	Nm
D02-27	Stuffing box flange, inner screws tightening torque	80	Nm
D02-28	Stuffing box halves, tightening torque	80	Nm
D02-29	Uppermost rings, ring-end clearance	4x6	mm
D02-30	Lowermost rings, ring-end clearance	3x3	mm
D02-33	Check length for the six uppermost springs:		
D02-34	– F 0 = 0 N................. L 0 =	639	mm
D02-35	– F 1 = 153 +/– 7 N....... L 1 =	763	mm
D02-36	– F 2 = 163 +/– 10 N..... L 2 =	835	mm
D02-37	Check length for the four lowermost springs:		
D02-38	– F 0 = 0 N................. L 0 =	496	mm
D02-39	– F 1 = 150 +/– 7 N....... L 1 =	646	mm
D02-40	– F 2 = 200 +/– 10 N..... L 2 =	714	mm
D02-44	Stuffing box complete	63	kg
D02-45	Stuffing box half	19	kg

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P**909**51 refers to chapter 909.

Plate	Item No.	Description
P90251	109	Mounting tool for stuffing box spring
P90251	122	Worktable for stuffing box
P90451	118	Rubber cover for crosshead

1. After the piston rod stuffing box has been dismantled, check the following clearances:.
   See Procedure 902-2.2.

   <u>Uppermost scraper ring and sealing rings</u>

   Clearance at ring ends (scraper ring).
   Total clearance (scraper ring). (D02-29)

   Clearance at ring ends (sealing rings).
   Total clearance (sealing rings). (D02-29)

   <u>Lowermost scraper rings</u>

   Clearance at ring ends.
   Total clearance. (D02-30)

   The ring clearances stated in Data apply to new rings.

   As a general guide, it is recommended – depending on the overhauling intervals and one's own experience – to replace sealing rings and scraper rings when the specified clearance values D02-29 and D02-30 have been halved.

2. Check the length of the springs at different loads. Adjust the loads to achieve the lengths stated in Data.

   If the loads necessary are outside the limits stated in Data, the springs must be discarded.

   It is recommended to renew the springs for the sealing rings when the rings are renewed.

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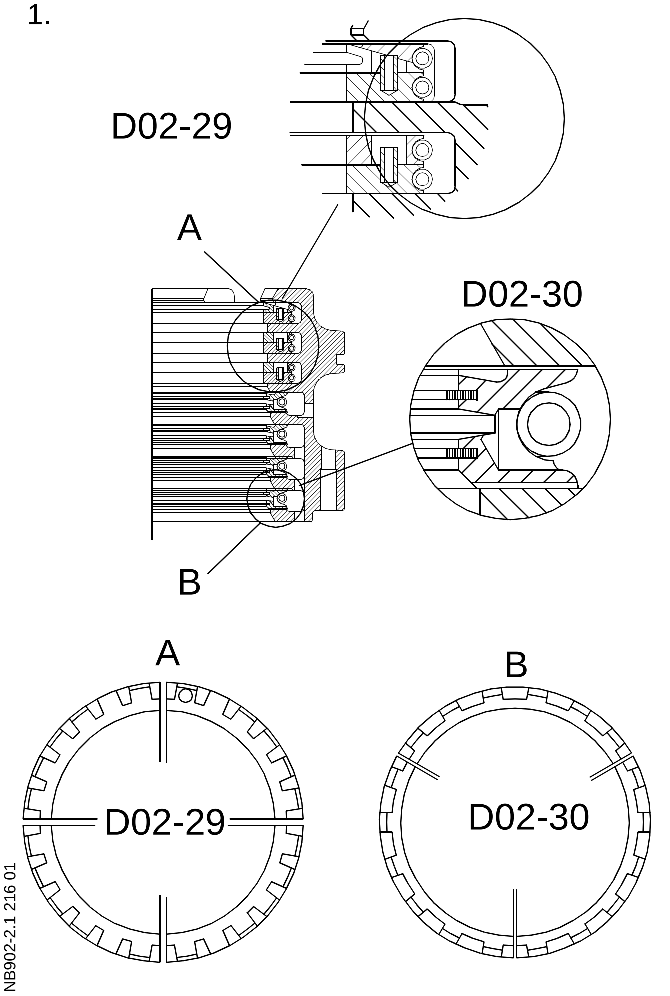

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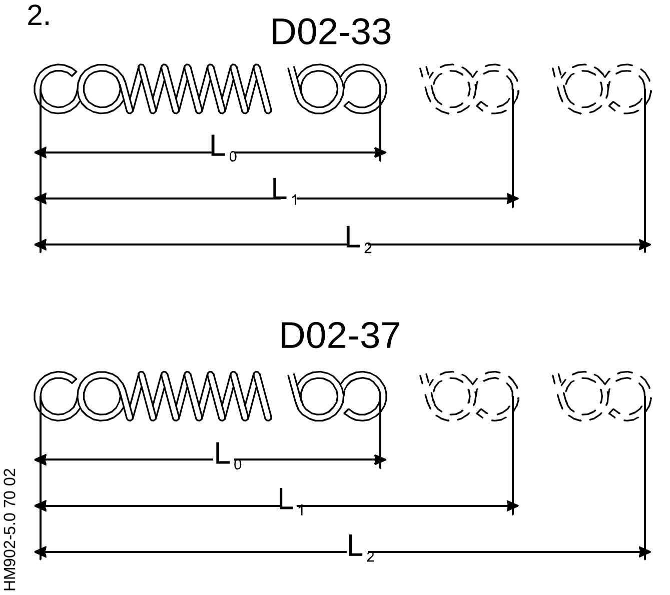

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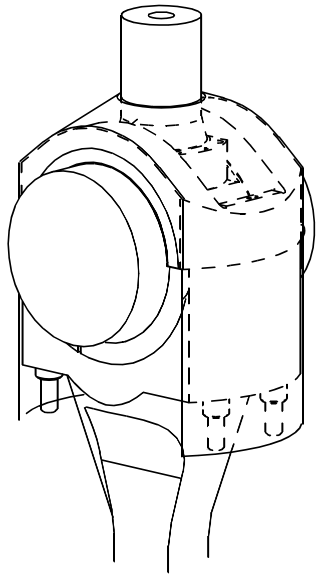

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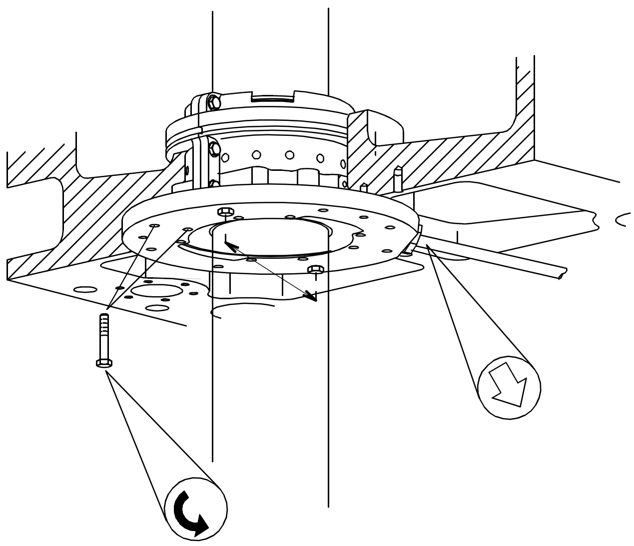

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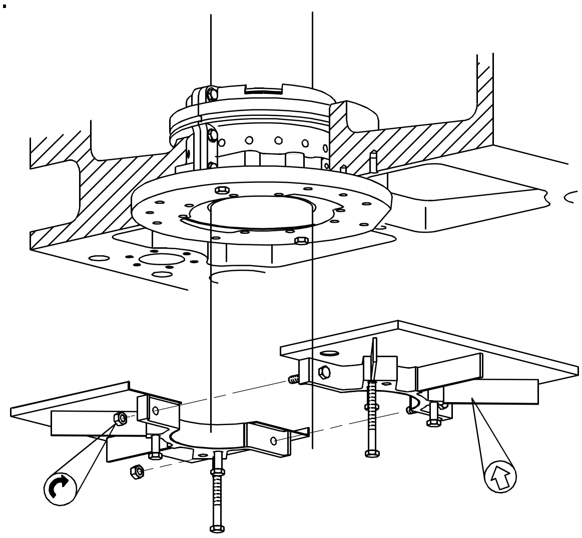

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In connection with dismantling of the piston, only the innermost screws in the stuffing box flange should be removed.

If, in the period between piston overhauls, it becomes necessary to inspect the piston rod stuffing box, proceed as follows:

1. Turn the crosshead to about 90° from TDC.

   Mount the rubber cover around the piston rod to protect the crosshead bearing from impurities.

2. Remove the drain oil pipe and all innermost screws and all outer screws except for two screws placed diametrically opposite in the stuffing box flange, longitudinally to the engine.

3. Mount the worktable around the piston rod so that the two remaining screws in the stuffing box flange can be loosened through the holes.

4. Remove the two long dismantling screws from the worktable.

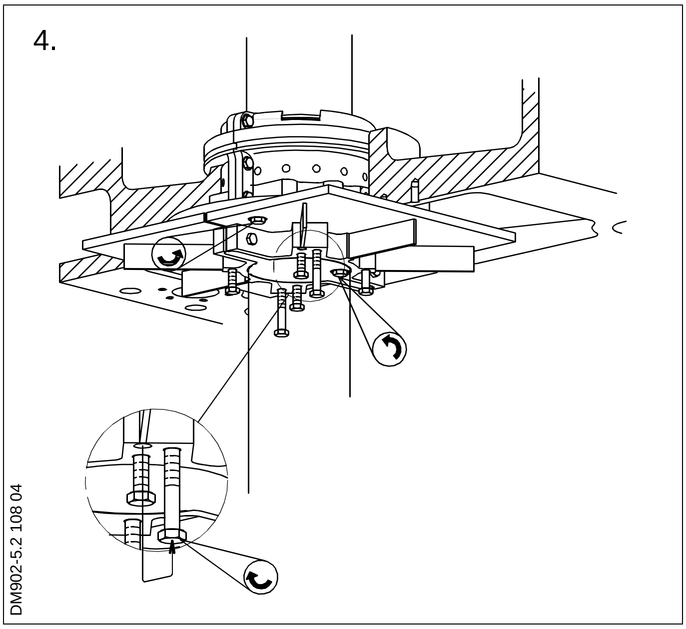
   Mount them in the stuffing box through the holes in the worktable.

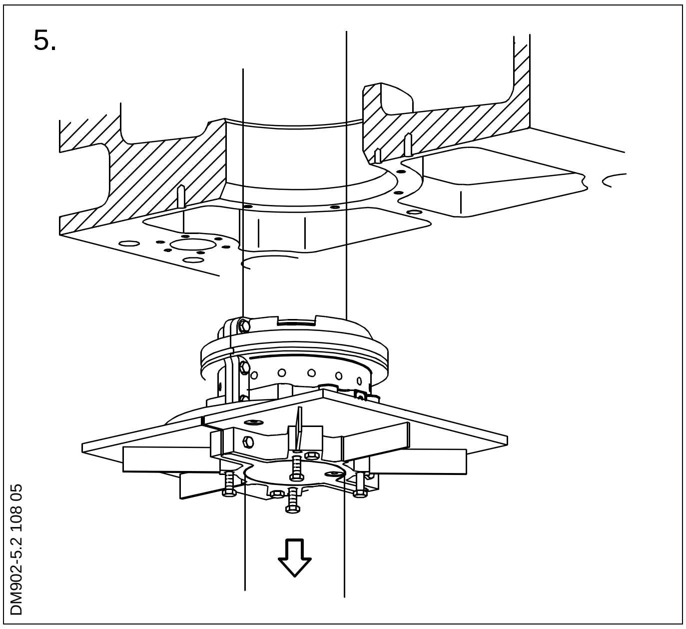
   Remove the remaining two screws from the stuffing box.

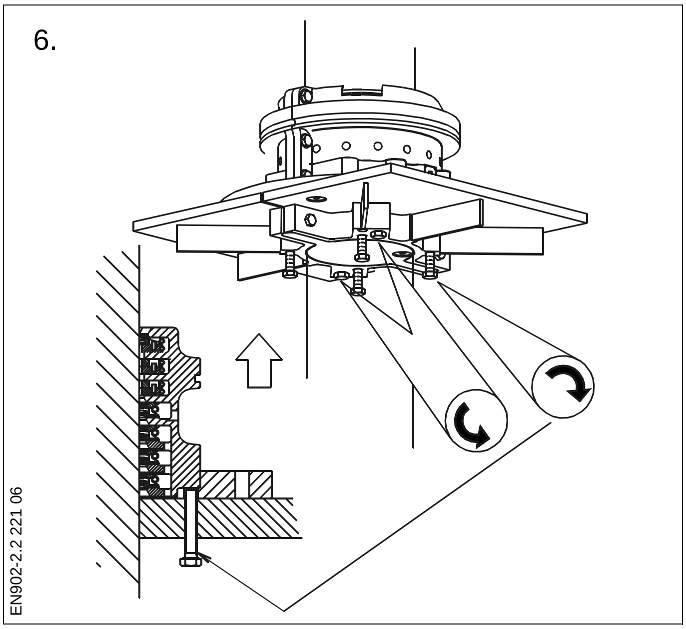
5. Turn the piston to BDC, thereby withdrawing the stuffing box from the cylinder frame bottom.

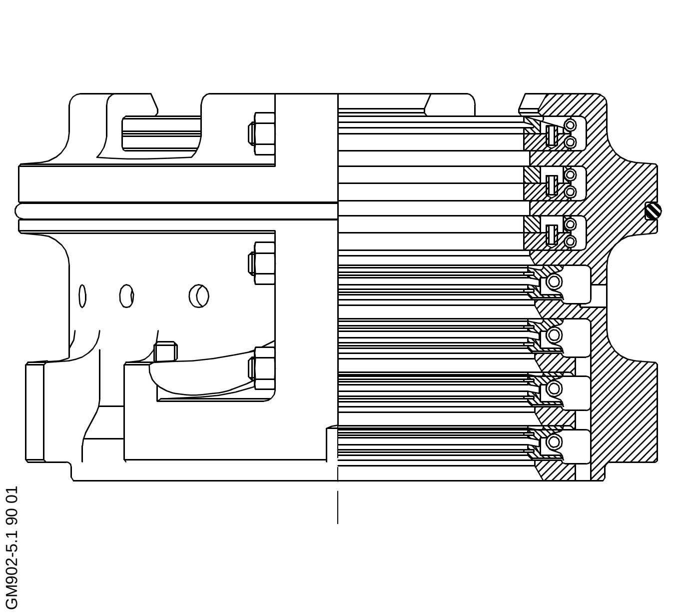
6. Remove the two long dismantling screws from the stuffing box and mount them in the worktable.

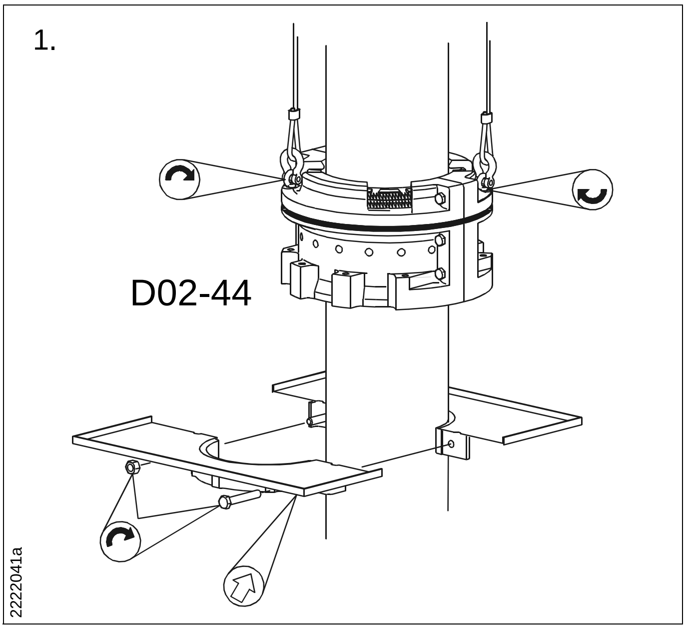
   By means of the four short screws in the worktable, press the stuffing box out of the flange.
   For overhauling the stuffing box, see Procedure 902-2.3.

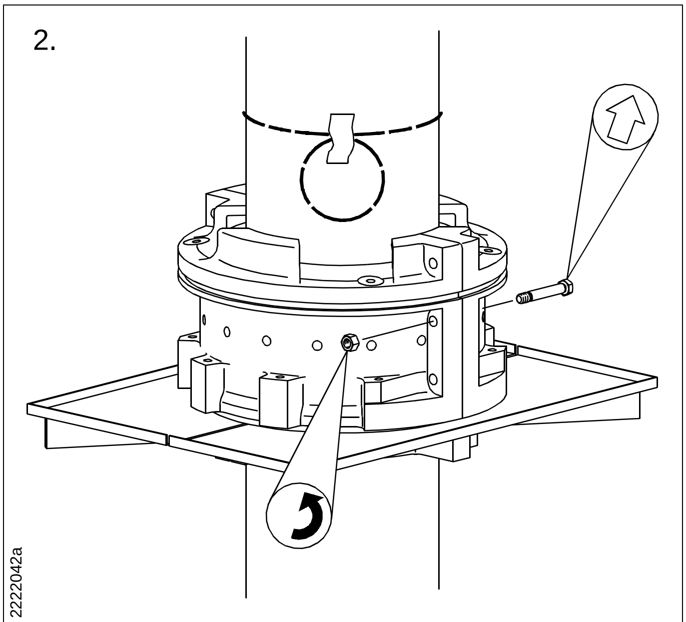












Normally, overhaul of the piston rod stuffing box is carried out by routine methods in connection with the dismantling (pulling) of the pistons.

During such overhauls, the piston rests on a support placed over one of the cut-outs in the top platform.

Work on the stuffing box is then carried out from the platform below.

Overhaul inside the engine is carried out in the same way as outside the engine.

1. Mount two eye bolts in the stuffing box flange, and hook on two tackles.

   Lift the stuffing box a little up the piston rod, and mount the worktable round the piston rod at a suitable working height.

   Land the stuffing box on the worktable, and remove the tackles and eye bolts.

2. Remove the O-ring of the stuffing box. If the O-ring is intact and is to be used again, move it up the piston rod and secure it in this position, for example with tape.

   Remove the nuts from the stuffing box assembling bolts.

3. Take out the six bolts, and pull away one stuffing box half.

   Mount two eye bolts on the stuffing box half and remove it from the worktable.

4. Using a feeler gauge, measure the vertical clearance of the rings.
   See Procedure 902-2.1.

5. Remove the remaining stuffing box half and press all sealing rings and scraper rings down against the worktable.

6. Measure the clearance between the ring segments to determine whether replacement is necessary.
   See Procedure 902-2.1.

   Dismantle and stack the rings in the same order as when fitted in the stuffing box.

   Carefully clean all the ring segments.

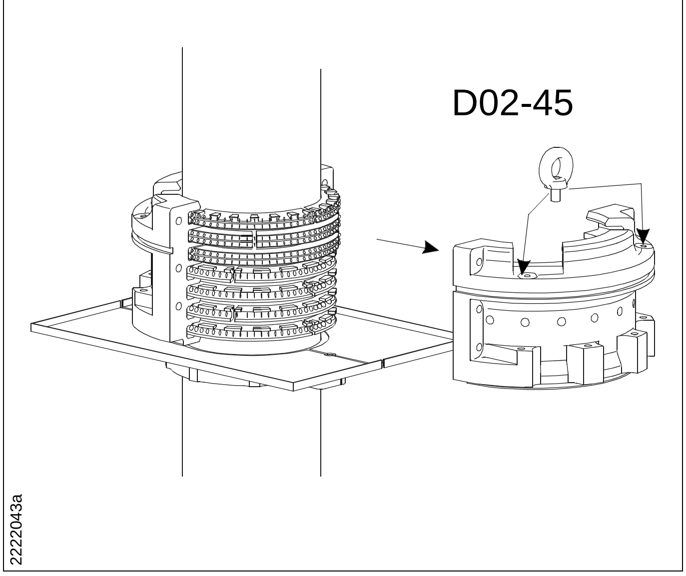
   Inspect and assess the surface quality of the sealing rings. If their sliding surfaces have scratches or marks, replace the rings.

7. Check the lengths of the springs.
   See Procedure 902-2.1.

8. Inspect the surface of the piston rod. If small longitudinal scratches have occurred (caused by poorly adapted stuffing box rings), smooth the piston rod surface carefully with a fine grained carborundum stone. In the case of coarse scratches, it may prove necessary to machine-grind the surface in a workshop.

9. Clean the halves of the stuffing box housing.

3.



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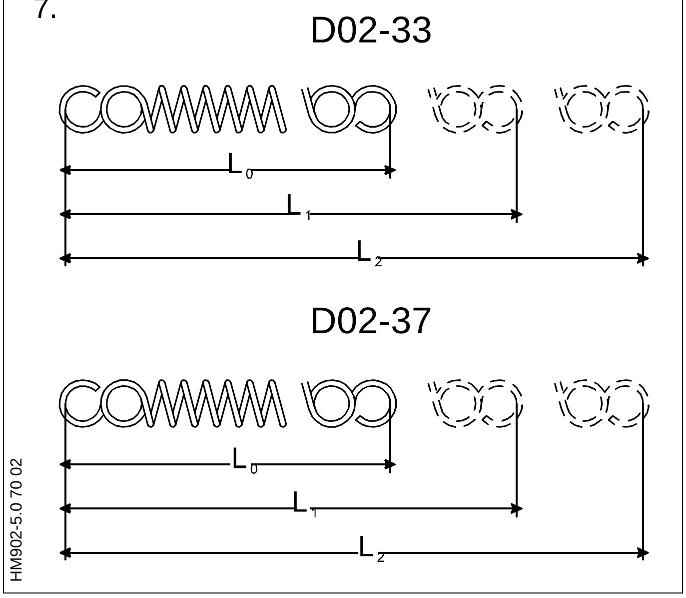

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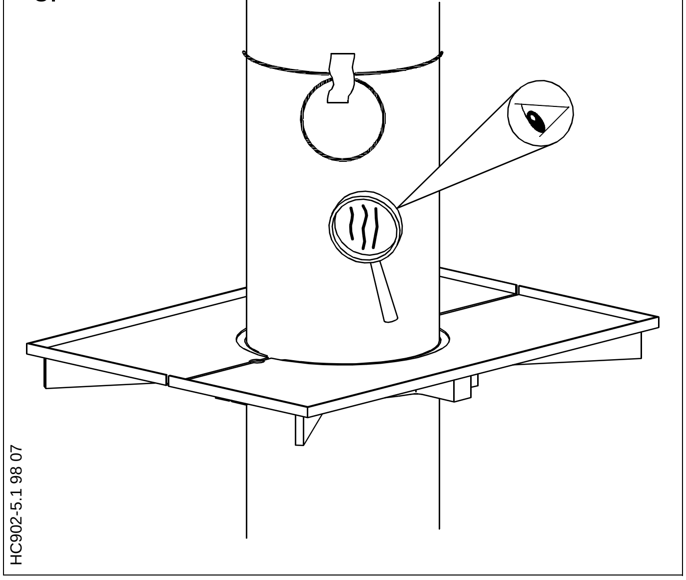

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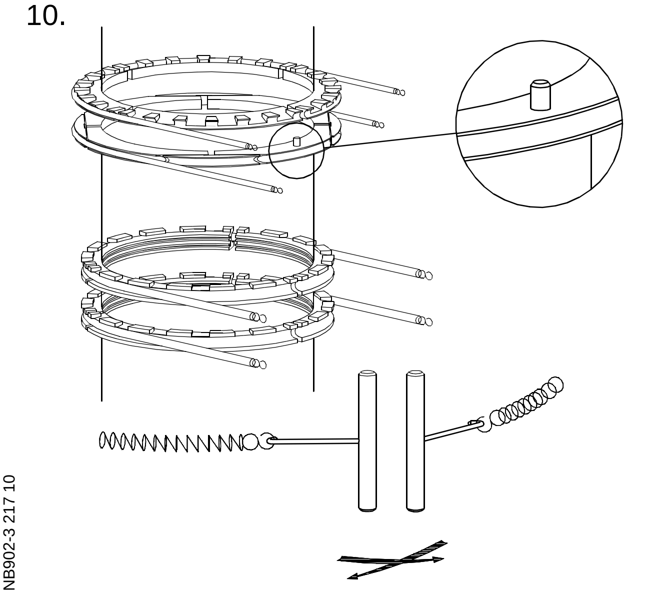

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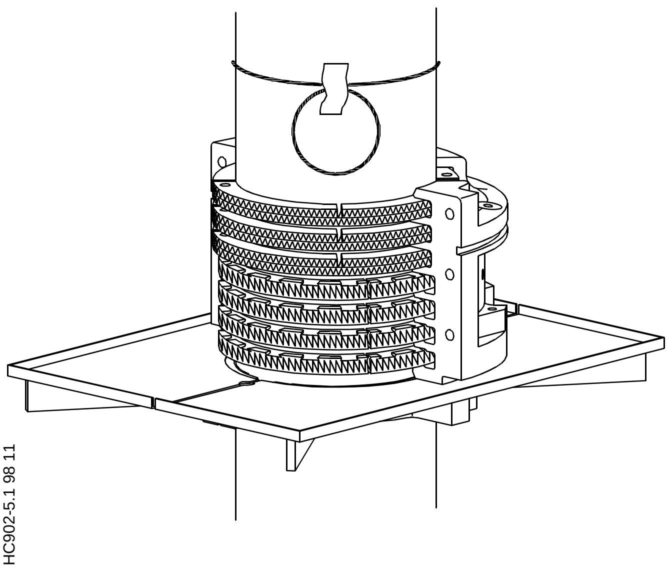


10. Lubricate the piston rod (in the area where all the ring units in the stuffing box will be positioned) with molybdenum disulphide (MoS_2).

    Assemble all the stuffing box ring units round the piston rod, on the worktable, in the following way:

    - Place the lowermost scraper ring segments on the worktable.
    - Place the spring round the segments, and hook the spring ends together.

    Repeat this procedure for the remaining scraper rings.

    On top of the scraper rings, assemble the two sealing ring units (each consisting of a 4-part and an 8-part ring).

    Assemble the 8-part sealing ring so that the two guide pins face upwards, place the spring round the segments and, hook the spring ends together.

    Assemble the 4-part sealing ring above the 8-part sealing ring. Push the two rings together in such a manner that the guide pins in the lower sealing ring engage with the two holes in the upper sealing ring.

    Finally, assemble the uppermost ring unit (consisting of a 4-part scraper ring and an 8-part sealing ring).

11. Use the stuffing box half on the worktable to adjust the height of all the assembled ring units on the piston rod until the ring units are opposite the corresponding grooves in the stuffing box housing. Subsequently, push the stuffing box half into contact with the piston rod, round the ring units.

Note!
If the stuffing box is assembled inside the engine, place two pieces of plywood of the same thickness as the flange on the worktable, to ease the assembling.

12. Check the ring clearance again.

    Then place the other half of the stuffing box housing on the worktable, pushing it into place round the rings.

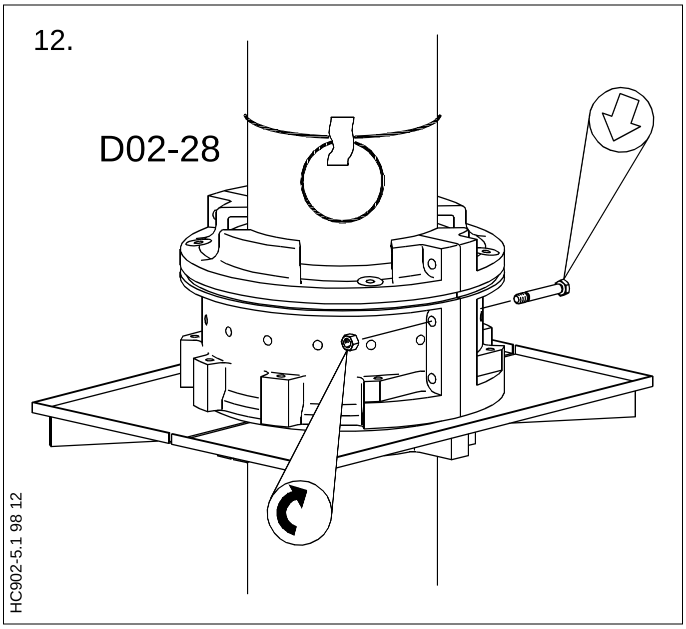
    Mount and tighten up the fitted bolts to the torque specified on the Data Sheet.

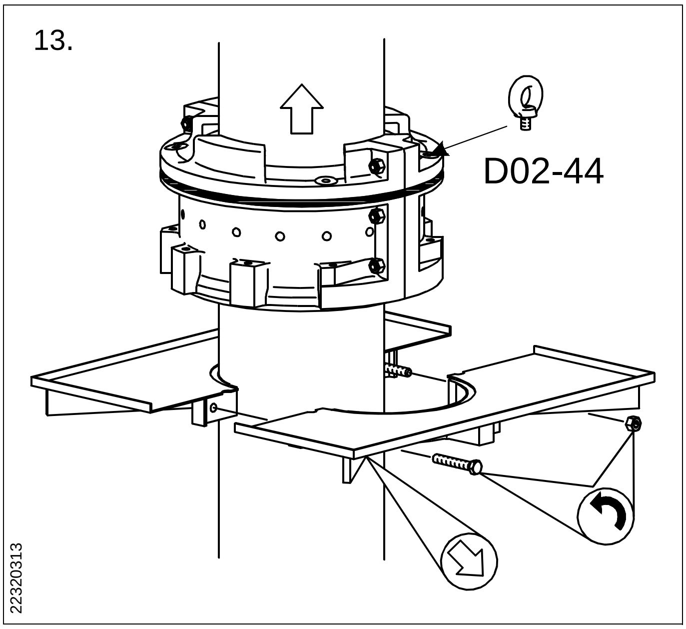
    Mount the O-ring in the stuffing box groove.

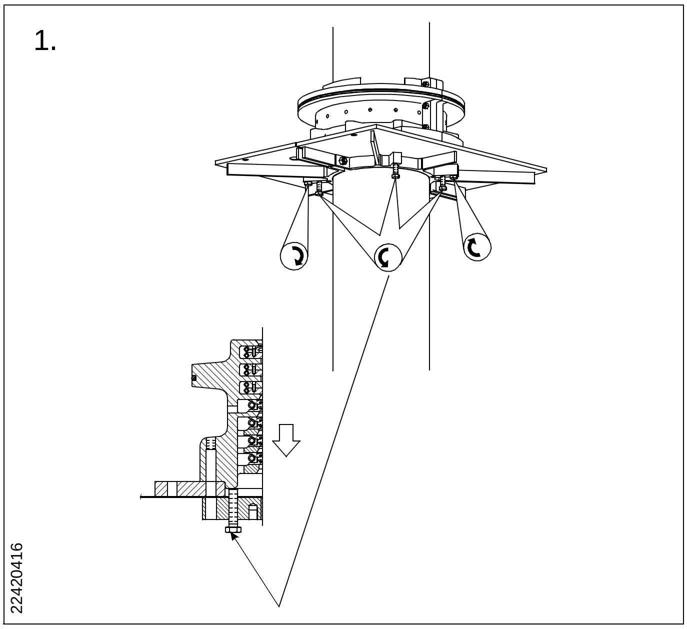
13. Mount eye bolts and wire ropes, and lift the stuffing box a little.

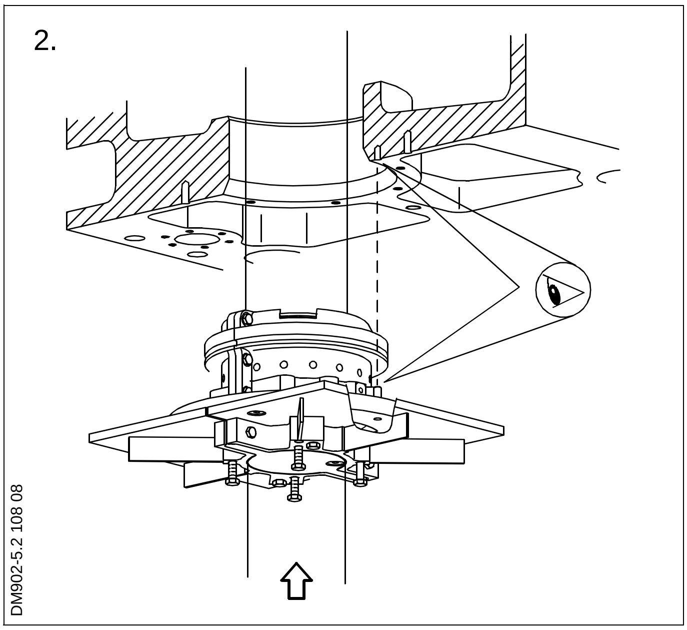
    Remove the worktable and lower the stuffing box until it rests against the distance pieces on the piston rod foot.

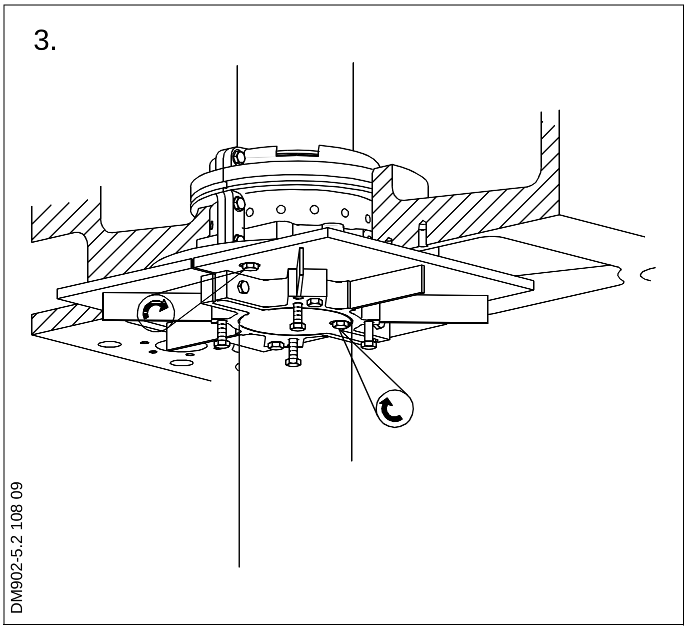
    Remove wire ropes and screws.











1. In connection with mounting of the piston, only the innermost flange screws are to be mounted and tightened, see data.

   After overhauling the stuffing box inside the engine, assemble the stuffing box halves on top of the four screws.

   Mount the two long screws from the worktable in the stuffing box.

   Turn down the short screws so that the stuffing box lands on the flange.

2. Turn the piston upwards until the stuffing box is in place in the cylinder frame.

Note!
Make sure that the two guide pins in the flange enter the guide holes in the bottom of the cylinder frame.

3. Mount two screws in the flange through the holes in the worktable.

4. Remove the long screws from the stuffing box and mount them in the worktable.

   Remove the worktable from the piston rod.

5. Mount and tighten all the inner and outer screws for the stuffing box. *See Data.*

6. Remove the protecting rubber cover from the piston rod/crosshead.

   Smear the piston rod with molybdenum disulphide.

   Then turn the crankshaft a couple of revolutions.

   Start up the engine and keep it running for about fifteen minutes at a number of revolutions corresponding to very slow or idle speed.

   Then stop the engine and inspect the piston rod and stuffing box.

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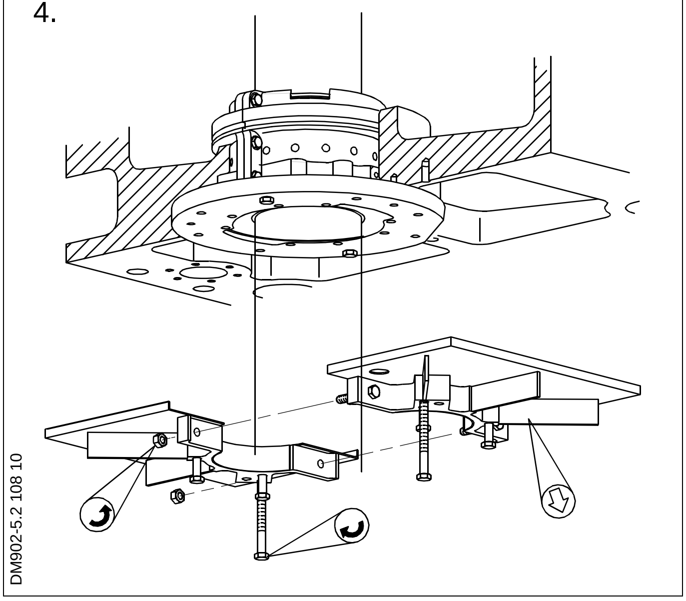

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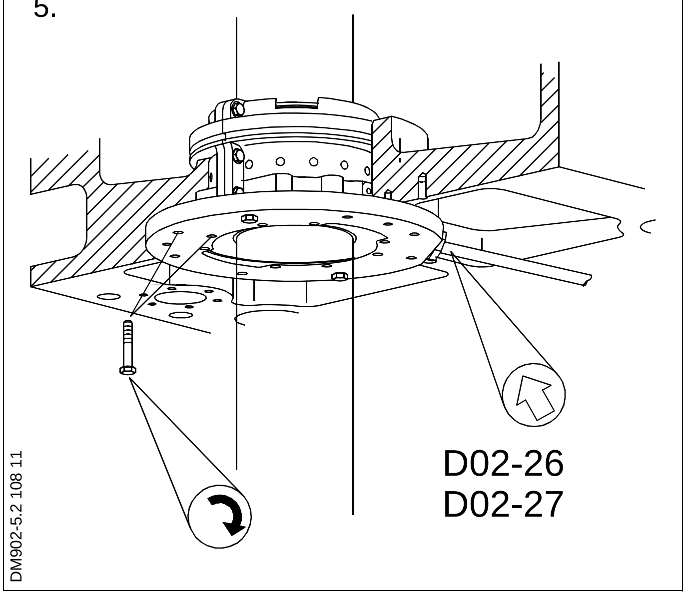

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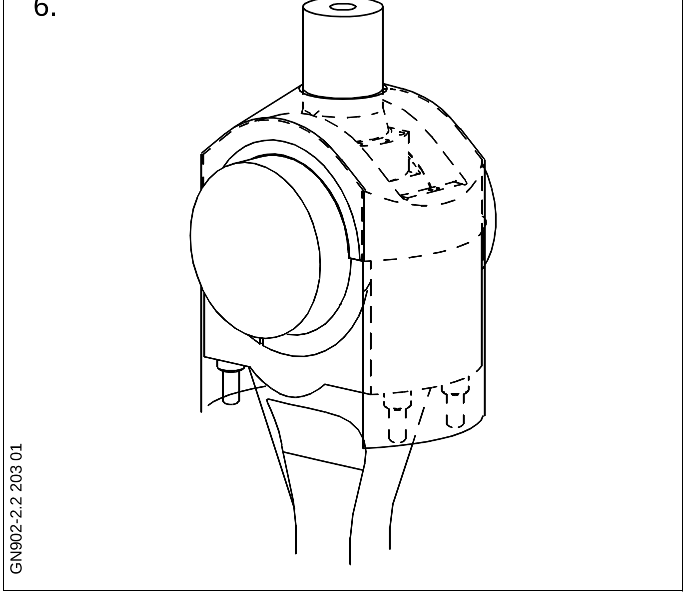
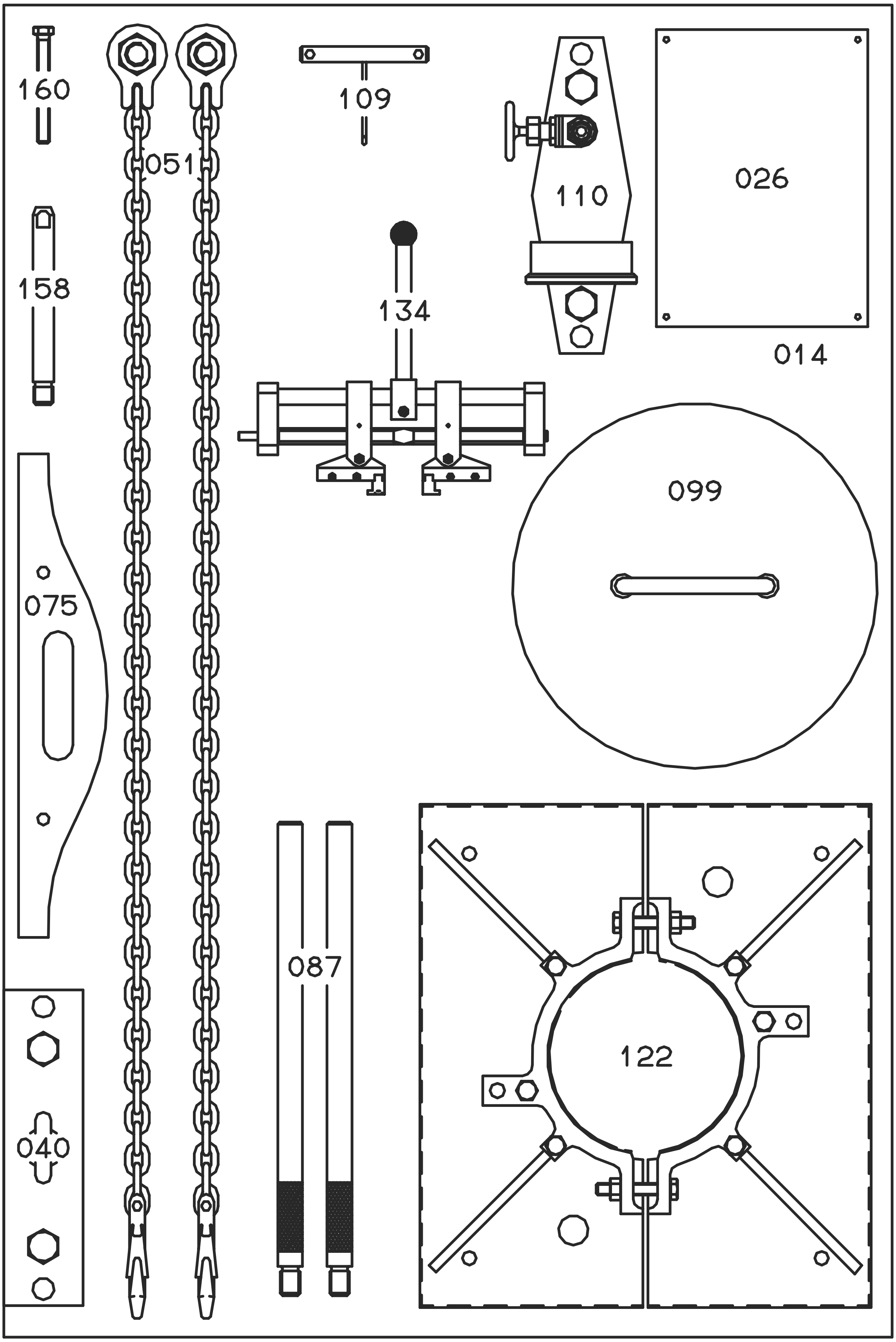
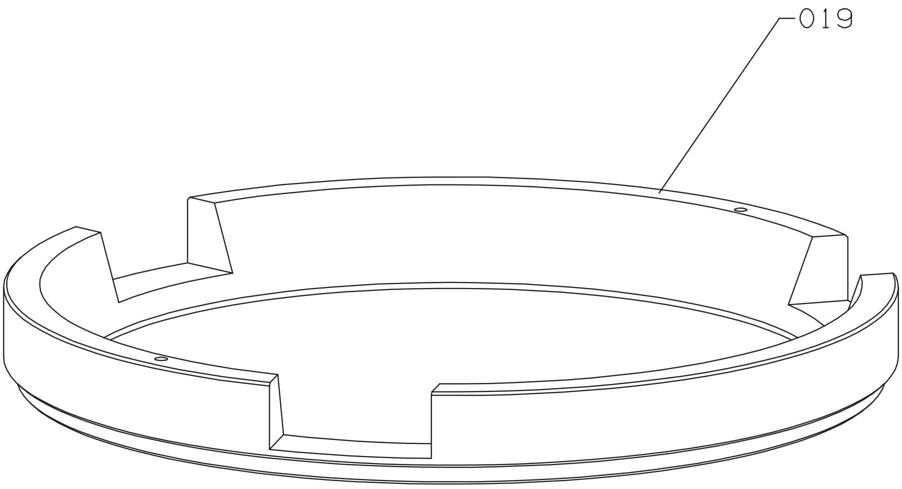



160
051
109
110
026
158
134
014
099
075
087
122
040

Piston and Piston Rod Panel

Item No.	Item Description
014	Panel for tools
026	Name plate
040	Lifting tool, piston rod foot
051	Lifting tool, cylinder liner
075	Template, piston top
087	Distance piece, stuffing box
099	Cover, stuffing box hole
109	Mounting tool, stuffing box spring
110	Pressure test tool, piston
122	Worktable, stuffing box
134	Expander, piston ring
158	Guide screw, piston crown
160	Guide screw, piston crown

Item No.	Item Description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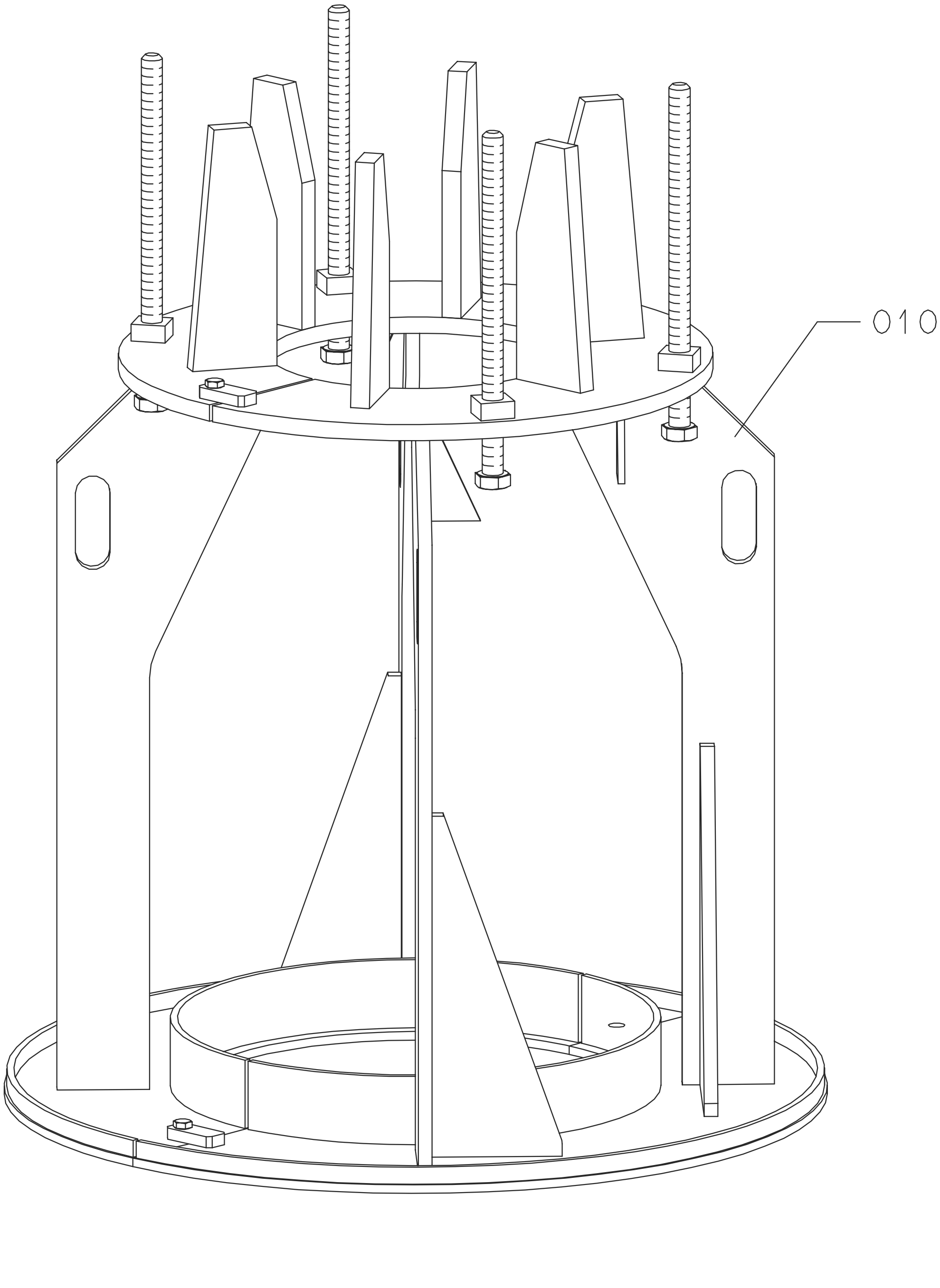
Piston and Piston Rod - Tools

Item No.	Item Description
019	Guide ring for piston

Item No.	Item Description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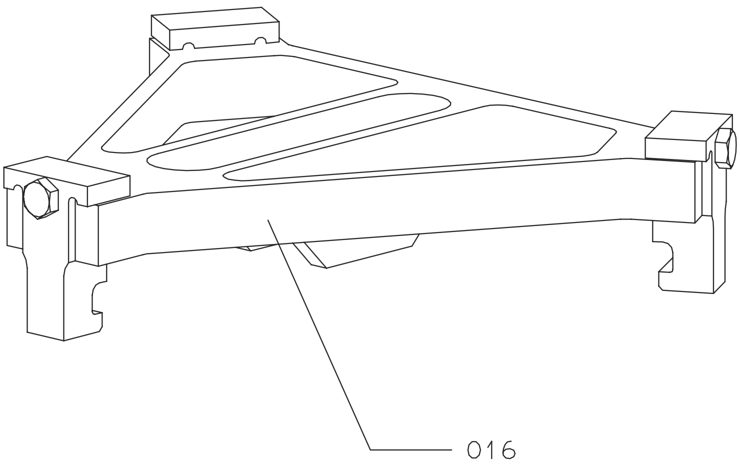
Piston and Piston Rod - Support Tools

Item No.	Item Description	Item No.	Item Description
010	Support iron for piston		


016

Piston - Lifting Tools

Item No.	Item Description
016	Lifting tool for piston

Item No.	Item Description